

PRAKTIKUM SISTEM OPERASI

MODUL 15

Keamanan Linux



Oleh:

Candra Dinata - 2311104061

PROGRAM STUDI S1 REKAYASA PERANGKAT LUNAK

DIREKTORAT KAMPUS PURWOKERTO

UNIVERSITAS TELKOM

2026

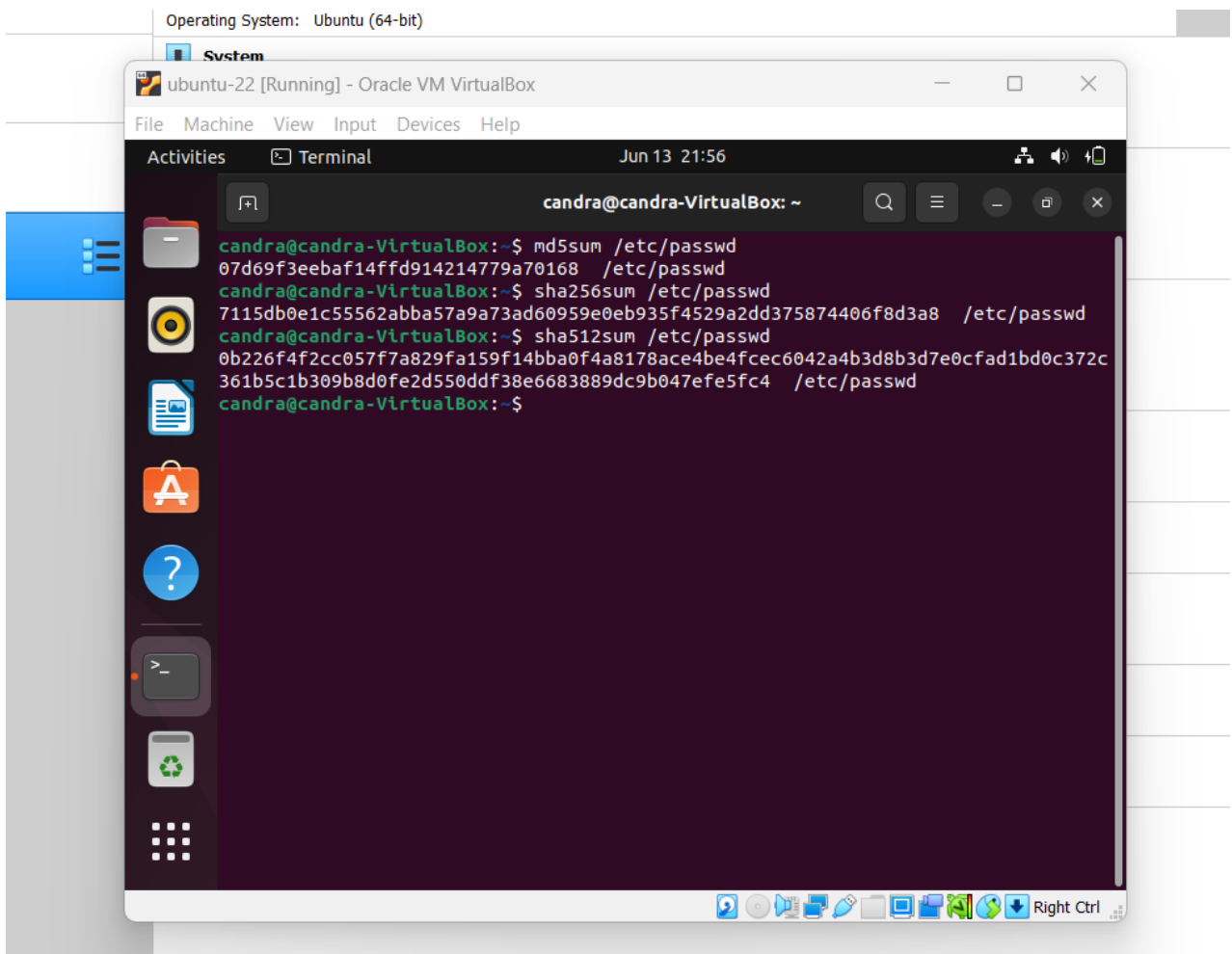
JURNAL

1. Integritas: Dasar Hashing

A. Hash file /etc/passwd

Jalankan:

```
md5sum /etc/passwd  
sha256sum /etc/passwd  
sha512sum /etc/passwd
```

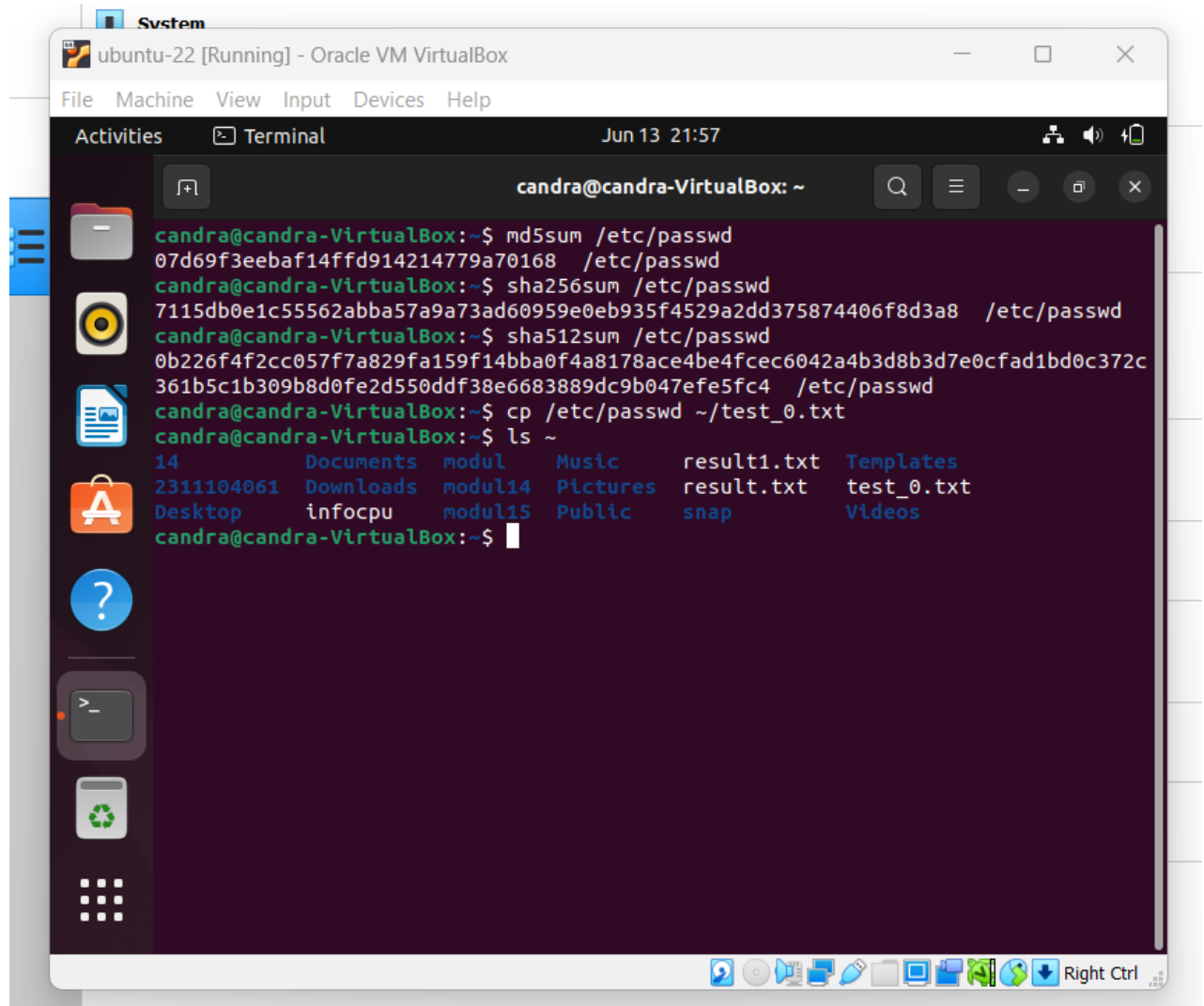


B. Buat salinan file

```
cp /etc/passwd ~/test_0.txt
```

Cek:

Is ~



The screenshot shows a terminal window titled 'candra@candra-VirtualBox: ~' with a dark purple background. The terminal displays the following commands and their outputs:

```
candra@candra-VirtualBox:~$ md5sum /etc/passwd
07d69f3eebaf14ffd914214779a70168 /etc/passwd
candra@candra-VirtualBox:~$ sha256sum /etc/passwd
7115db0e1c55562abba57a9a73ad60959e0eb935f4529a2dd375874406f8d3a8 /etc/passwd
candra@candra-VirtualBox:~$ sha512sum /etc/passwd
0b226f4f2cc057f7a829fa159f14bba0f4a8178ace4be4fcec6042a4b3d8b3d7e0cfad1bd0c372c
361b5c1b309b8d0fe2d550ddf38e6683889dc9b047efe5fc4 /etc/passwd
candra@candra-VirtualBox:~$ cp /etc/passwd ~/test_0.txt
candra@candra-VirtualBox:~$ ls ~
14          Documents  modul    Music     result1.txt  Templates
2311104061  Downloads  modul14  Pictures  result.txt   test_0.txt
Desktop     infocpu    modul15  Public   snap         Videos
```

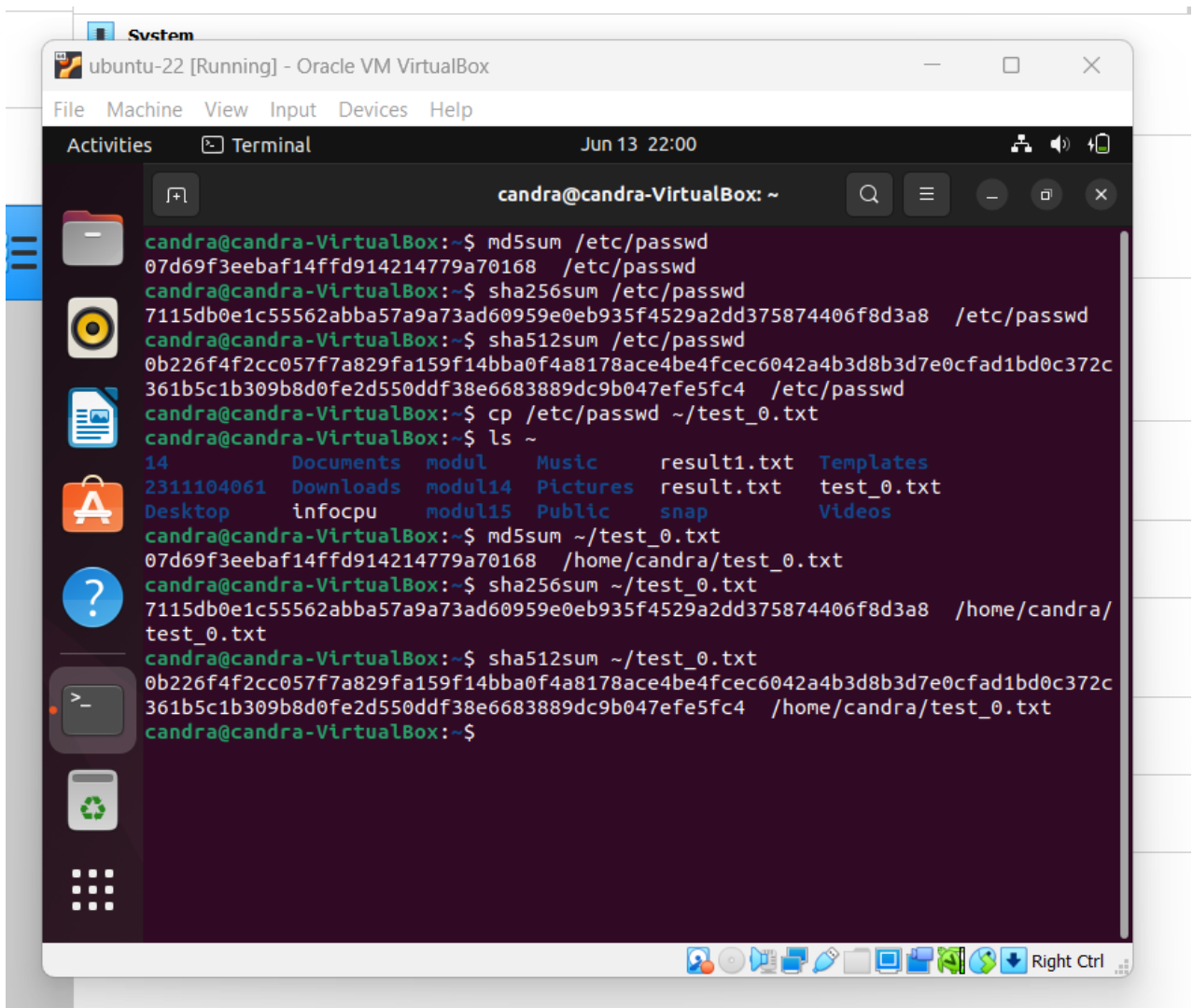
The terminal window is part of an Oracle VM VirtualBox interface, with a menu bar (File, Machine, View, Input, Devices, Help) and a toolbar (Activities, Terminal, Jun 13 21:57). The system is identified as 'ubuntu-22 [Running] - Oracle VM VirtualBox'.

C. Hash file test_0.txt

md5sum ~/test_0.txt

sha256sum ~/test_0.txt

sha512sum ~/test_0.txt



D. Rename file

`mv ~/test_0.txt ~/file_0.txt`

Hash kembali:

`md5sum ~/file_0.txt`

`sha256sum ~/file_0.txt`

`sha512sum ~/file_0.txt`

The screenshot shows a terminal window titled 'candra@candra-VirtualBox: ~' with a dark background. The user is performing several operations to verify file integrity and demonstrate that renaming a file does not change its hash. The commands and their outputs are as follows:

```
candra@candra-VirtualBox:~$ sha256sum /etc/passwd
7115db0e1c55562abba57a9a73ad60959e0eb935f4529a2dd375874406f8d3a8 /etc/passwd
candra@candra-VirtualBox:~$ sha512sum /etc/passwd
0b226f4f2cc057f7a829fa159f14bba0f4a8178ace4be4fcec6042a4b3d8b3d7e0cfad1bd0c372c361b5c1b309b8d0fe2d550ddf38e6683889dc9b047efe5fc4 /etc/passwd
candra@candra-VirtualBox:~$ cp /etc/passwd ~/test_0.txt
candra@candra-VirtualBox:~$ ls ~
14 Documents modul Music result1.txt Templates
2311104061 Downloads modul14 Pictures result.txt test_0.txt
Desktop infocpu modul15 Public snap Videos
candra@candra-VirtualBox:~$ md5sum ~/test_0.txt
07d69f3eebaf14ffd914214779a70168 /home/candra/test_0.txt
candra@candra-VirtualBox:~$ sha256sum ~/test_0.txt
7115db0e1c55562abba57a9a73ad60959e0eb935f4529a2dd375874406f8d3a8 /home/candra/test_0.txt
candra@candra-VirtualBox:~$ sha512sum ~/test_0.txt
0b226f4f2cc057f7a829fa159f14bba0f4a8178ace4be4fcec6042a4b3d8b3d7e0cfad1bd0c372c361b5c1b309b8d0fe2d550ddf38e6683889dc9b047efe5fc4 /home/candra/test_0.txt
candra@candra-VirtualBox:~$ mv ~/test_0.txt ~/file_0.txt
candra@candra-VirtualBox:~$ md5sum ~/file_0.txt
07d69f3eebaf14ffd914214779a70168 /home/candra/file_0.txt
candra@candra-VirtualBox:~$ sha256sum ~/file_0.txt
7115db0e1c55562abba57a9a73ad60959e0eb935f4529a2dd375874406f8d3a8 /home/candra/file_0.txt
candra@candra-VirtualBox:~$ sha512sum ~/file_0.txt
0b226f4f2cc057f7a829fa159f14bba0f4a8178ace4be4fcec6042a4b3d8b3d7e0cfad1bd0c372c361b5c1b309b8d0fe2d550ddf38e6683889dc9b047efe5fc4 /home/candra/file_0.txt
candra@candra-VirtualBox:~$
```

The terminal window is part of an Oracle VM VirtualBox interface, with the title bar showing 'ubuntu-22 [Running] - Oracle VM VirtualBox'. The interface includes a menu bar (File, Machine, View, Input, Devices, Help) and a toolbar with icons for Activities, Terminal, and search. The system clock shows 'Jun 13 22:02'.

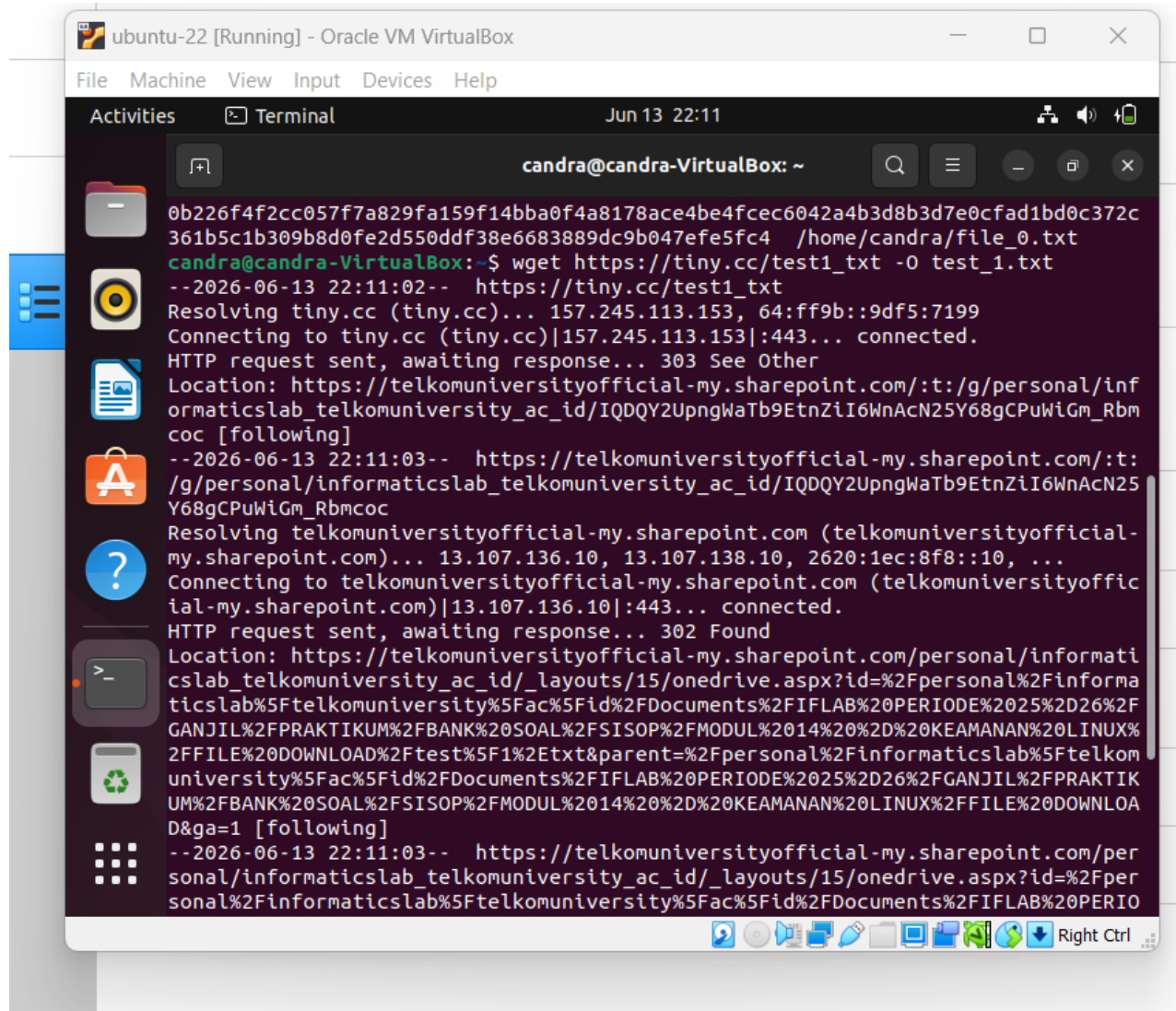
E. Analisis

Hash ketiga file sama karena isi file identik. Perubahan nama file (rename) tidak mengubah isi file sehingga hash tetap sama.

2. Avalanche Effect

A. Download test_1.txt

wget https://tiny.cc/test1_txt -O test_1.txt



The screenshot shows a terminal window titled 'candra@candra-VirtualBox: ~' with the following output for the command `wget https://tiny.cc/test1_txt -O test_1.txt`:

```
0b226f4f2cc057f7a829fa159f14bba0f4a8178ace4be4fcec6042a4b3d8b3d7e0cfad1bd0c372c
361b5c1b309b8d0fe2d550ddf38e6683889dc9b047efe5fc4 /home/candra/file_0.txt
candra@candra-VirtualBox:~$ wget https://tiny.cc/test1_txt -O test_1.txt
--2026-06-13 22:11:02-- https://tiny.cc/test1_txt
Resolving tiny.cc (tiny.cc)... 157.245.113.153, 64:ff9b::9df5:7199
Connecting to tiny.cc (tiny.cc)|157.245.113.153|:443... connected.
HTTP request sent, awaiting response... 303 See Other
Location: https://telkomuniversityofficial-my.sharepoint.com/:t:g/personal/informatics
lab_telkomuniversity_ac_id/IQDQY2UpngWaTb9EtnZiI6WnAcN25Y68gCPuWiGm_Rbm
coc [following]
--2026-06-13 22:11:03-- https://telkomuniversityofficial-my.sharepoint.com/:t:
/g/personal/informatics
lab_telkomuniversity_ac_id/IQDQY2UpngWaTb9EtnZiI6WnAcN25
Y68gCPuWiGm_Rbm
coc
Resolving telkomuniversityofficial-my.sharepoint.com (telkomuniversityofficial-
my.sharepoint.com)... 13.107.136.10, 13.107.138.10, 2620:1ec:8f8::10, ...
Connecting to telkomuniversityofficial-my.sharepoint.com (telkomuniversityoffic
ial-my.sharepoint.com)|13.107.136.10|:443... connected.
HTTP request sent, awaiting response... 302 Found
Location: https://telkomuniversityofficial-my.sharepoint.com/personal/informati
cslab_telkomuniversity_ac_id/_layouts/15/onedrive.aspx?id=%2Fpersonal%2Finforma
ticslab%5Ftelkomuniversity%5Fac%5Fid%2FDocuments%2FIFLAB%20PERIODE%2025%2D26%2F
GANJIL%2FPRAKTIKUM%2FBANK%20SOAL%2FSISOP%2FMODUL%2014%20%2D%20KEAMANAN%20LINUX%
2FFILE%20DOWNLOAD%2Ftest%5F1%2Etet&parent=%2Fpersonal%2Finformatics
lab%5Ftelkomuniversity%5Fac%5Fid%2FDocuments%2FIFLAB%20PERIODE%2025%2D26%2F
GANJIL%2FPRAKTIKUM%2FBANK%20SOAL%2FSISOP%2FMODUL%2014%20%2D%20KEAMANAN%20LINUX%2FFILE%20DOWNLOA
D&ga=1 [following]
--2026-06-13 22:11:03-- https://telkomuniversityofficial-my.sharepoint.com/per
sonal/informatics
lab_telkomuniversity_ac_id/_layouts/15/onedrive.aspx?id=%2Fper
sonal%2Finformatics
lab%5Ftelkomuniversity%5Fac%5Fid%2FDocuments%2FIFLAB%20PERIO
```

B. Hash sebelum diubah

md5sum ~/test_1.txt

sha256sum ~/test_1.txt

sha512sum ~/test_1.txt

The screenshot shows a terminal window titled 'candra@candra-VirtualBox: ~' with a dark background. The terminal output shows a file being downloaded from a SharePoint URL. The download progress bar indicates a size of 383,96K and a speed of 2,32MB/s. After the download, the user runs three commands to calculate the MD5, SHA256, and SHA512 hashes of the file 'test_1.txt'.

```
university%5Fac%5Fid%2FDocuments%2FIFLAB%20PERIODE%2025%2D26%2FGANJIL%2FPRAKTIK
UM%2FBANK%20SOAL%2FSISOP%2FMODUL%2014%20%2D%20KEAMANAN%20LINUX%2FFILE%20DOWNLOA
D&ga=1 [following]
--2026-06-13 22:11:03-- https://telkomuniversityofficial-my.sharepoint.com/per
sonal/informaticslab_telkomuniversity_ac_id/_layouts/15/onedrive.aspx?id=%2Fper
sonal%2Finformaticslab%5Ftelkomuniversity%5Fac%5Fid%2FDocuments%2FIFLAB%20PERIO
DE%2025%2D26%2FGANJIL%2FPRAKTIKUM%2FBANK%20SOAL%2FSISOP%2FMODUL%2014%20%2D%20KE
AMANAN%20LINUX%2FFILE%20DOWNLOAD%2Ftest%5F1%2Etxt&parent=%2Fpersonal%2Finformat
icslab%5Ftelkomuniversity%5Fac%5Fid%2FDocuments%2FIFLAB%20PERIODE%2025%2D26%2FG
ANJIL%2FPRAKTIKUM%2FBANK%20SOAL%2FSISOP%2FMODUL%2014%20%2D%20KEAMANAN%20LINUX%2
FFILE%20DOWNLOAD&ga=1
Reusing existing connection to telkomuniversityofficial-my.sharepoint.com:443.
HTTP request sent, awaiting response... 200 OK
Length: unspecified [text/html]
Saving to: 'test_1.txt'

test_1.txt          [ <=>          ] 383,96K  2,32MB/s   in 0,2s

2026-06-13 22:11:04 (2,32 MB/s) - 'test_1.txt' saved [393170]

candra@candra-VirtualBox:~$ md5sum ~/test_1.txt
387388e8a982045ce86bee182ddf2e07  /home/candra/test_1.txt
candra@candra-VirtualBox:~$ sha256sum ~/test_1.txt
0b198f642520a44b64bc8bc98d07fd7bac00aac31652f96e81f636ae8c8497f0  /home/candra/
test_1.txt
candra@candra-VirtualBox:~$ sha512sum ~/test_1.txt
2265f4ac416c3f6ea7741c4315304bce7b02949cd08a1ce364bb10db981b65dc7568da429b6f475
eb50599d28b2ac67fece2154adf8e1d80872726c2ac8168d6  /home/candra/test_1.txt
candra@candra-VirtualBox:~$
```

C. Hapus titik di akhir file

Lihat isi akhir file:

tail ~/test_1.txt

Buka editor:

nano ~/test_1.txt

Langkah:

1. Pergi ke akhir teks
2. Hapus tanda titik (.)

3. Simpan

Tekan:

Ctrl + X

Y

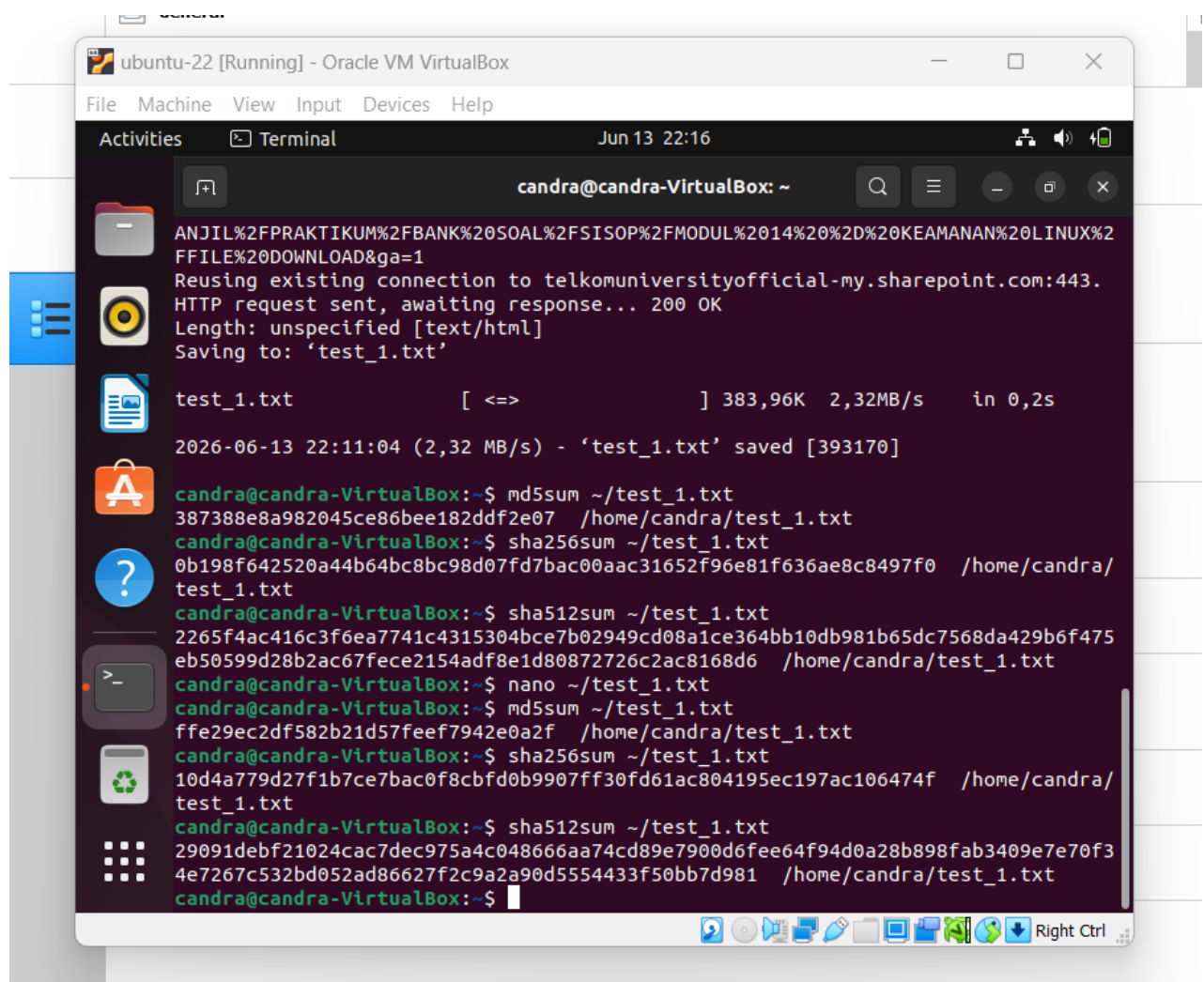
Enter

D. Hash kembali

md5sum ~/test_1.txt

sha256sum ~/test_1.txt

sha512sum ~/test_1.txt



E. Analisis

Walaupun hanya satu karakter yang dihapus, seluruh nilai hash berubah. Hal ini menunjukkan sifat Avalanche Effect, yaitu perubahan kecil pada data menghasilkan hash yang sangat berbeda.

3. Metadata File DOC

A. Download test_1.doc

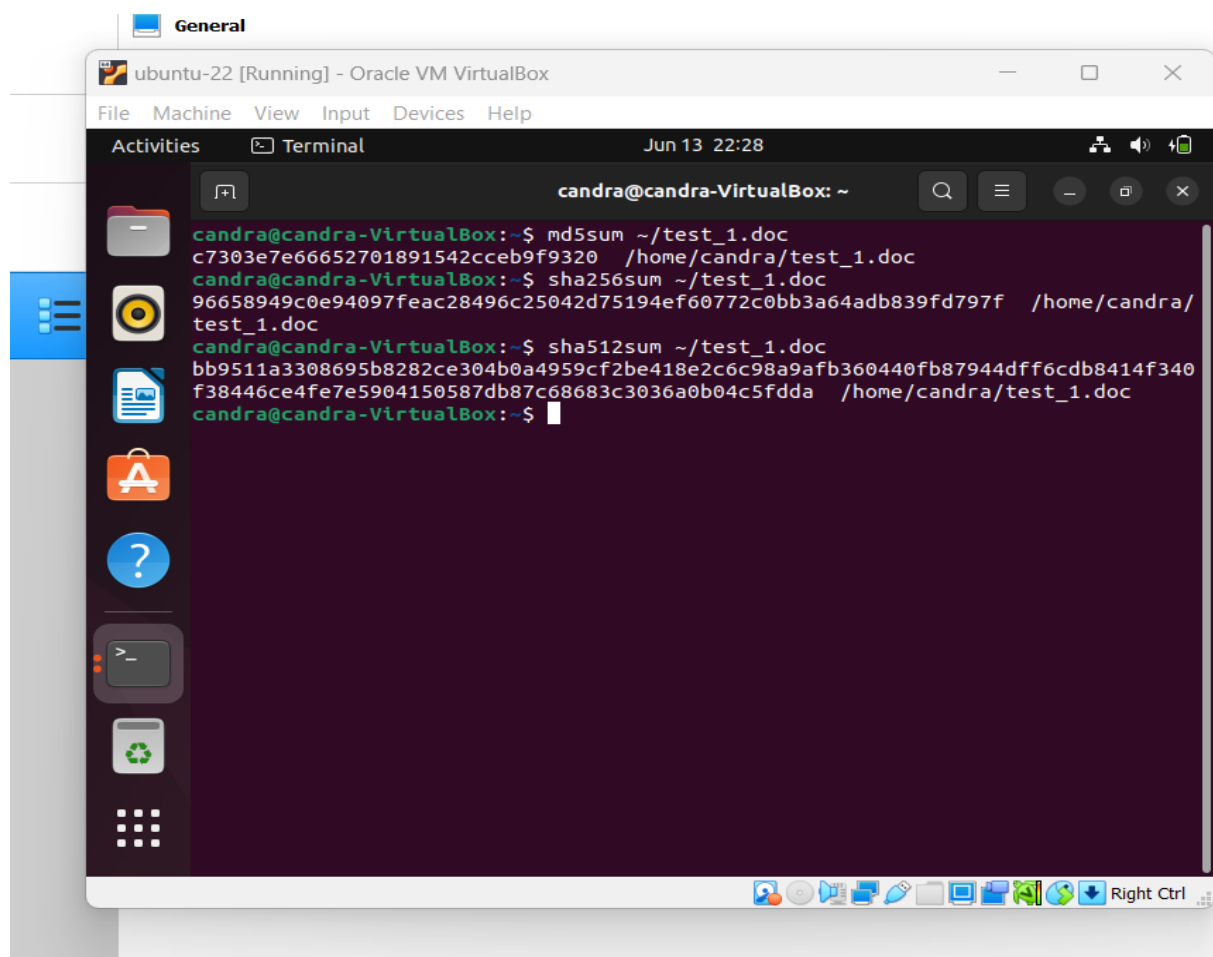
Download dari LMS atau sumber yang diberikan asisten.

B. Hash awal

```
md5sum ~/test_1.doc
```

```
sha256sum ~/test_1.doc
```

```
sha512sum ~/test_1.doc
```



The screenshot shows a terminal window titled 'ubuntu-22 [Running] - Oracle VM VirtualBox'. The terminal output is as follows:

```
candra@candra-VirtualBox: ~  
candra@candra-VirtualBox:~$ md5sum ~/test_1.doc  
c7303e7e66652701891542cceb9f9320  /home/candra/test_1.doc  
candra@candra-VirtualBox:~$ sha256sum ~/test_1.doc  
96658949c0e94097feac28496c25042d75194ef60772c0bb3a64adb839fd797f  /home/candra/  
test_1.doc  
candra@candra-VirtualBox:~$ sha512sum ~/test_1.doc  
bb9511a3308695b8282ce304b0a4959cf2be418e2c6c98a9afb360440fb87944dff6cdb8414f340  
f38446ce4fe7e5904150587db87c68683c3036a0b04c5fdda  /home/candra/test_1.doc  
candra@candra-VirtualBox:~$
```

C. Edit file

Buka LibreOffice:

libreoffice ~/test_1.doc

Tambahkan:

abcdef

Simpan:

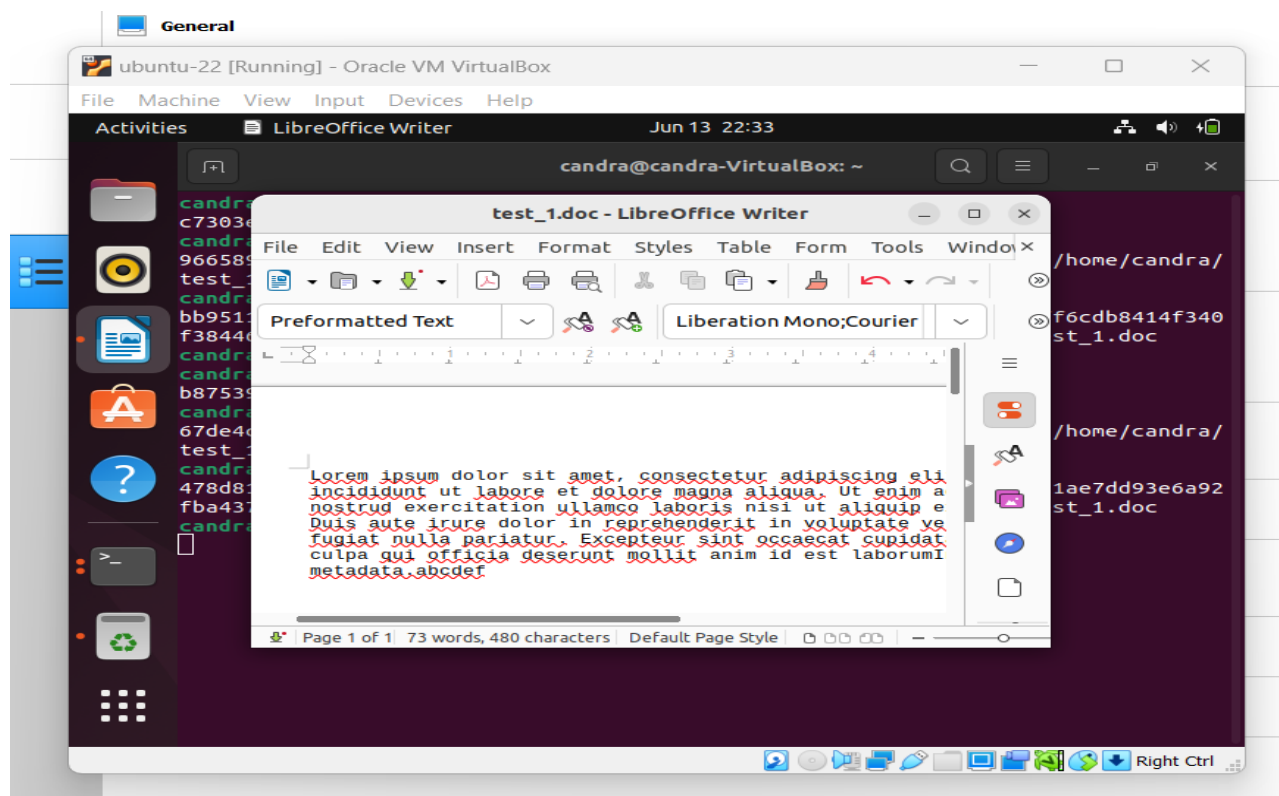
Ctrl + S

Pilih:

Keep Current Format

Kemudian:

1. Hapus tulisan abcdef
2. Simpan lagi

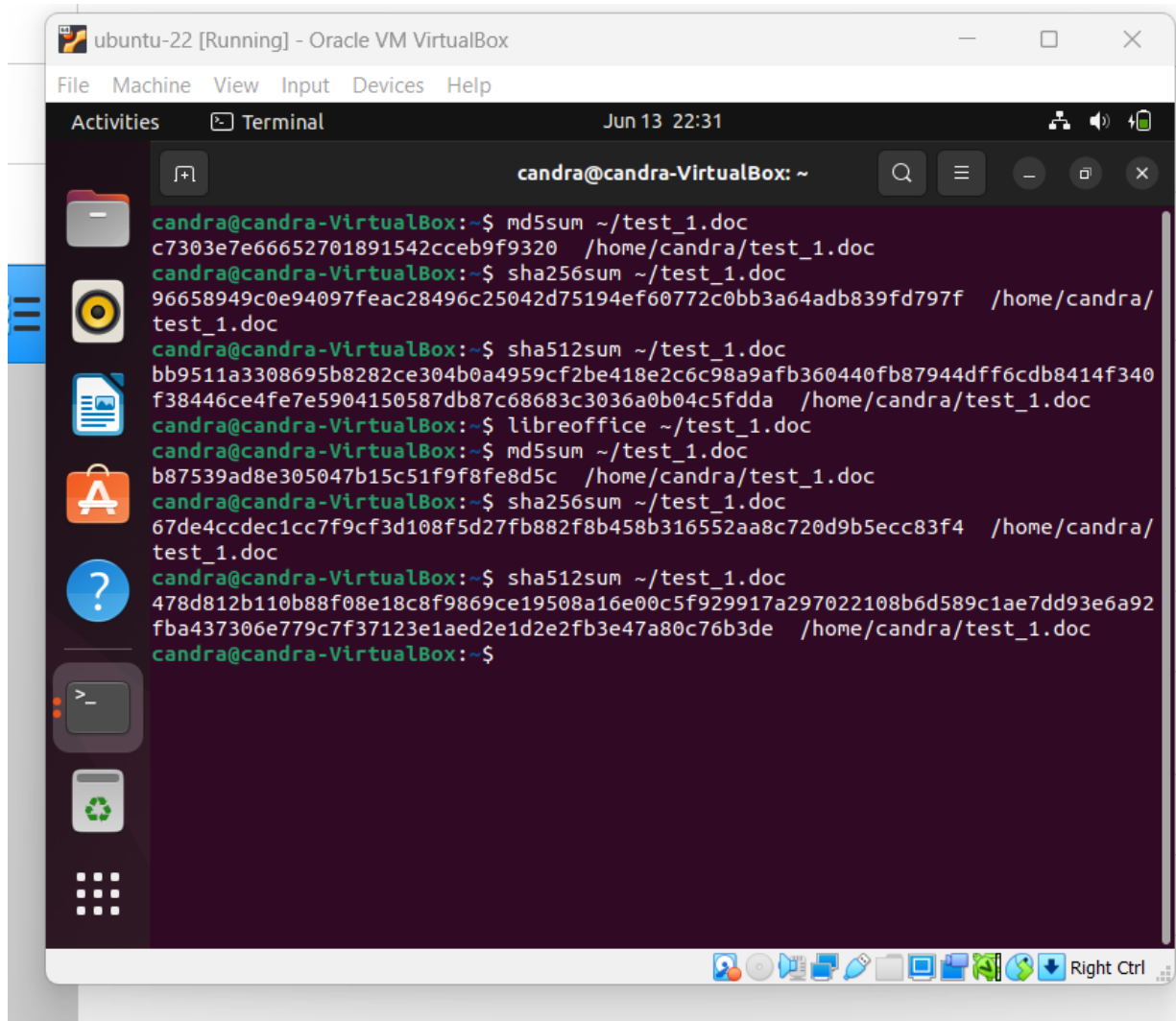


D. Hash ulang

md5sum ~/test_1.doc

sha256sum ~/test_1.doc

sha512sum ~/test_1.doc



The screenshot shows a terminal window titled 'ubuntu-22 [Running] - Oracle VM VirtualBox'. The window has a menu bar with 'File', 'Machine', 'View', 'Input', 'Devices', and 'Help'. Below the menu bar is a toolbar with 'Activities', 'Terminal', and a clock showing 'Jun 13 22:31'. The terminal itself has a title bar 'candra@candra-VirtualBox: ~' and a search bar. The terminal output shows the following commands and their results:

```
candra@candra-VirtualBox:~$ md5sum ~/test_1.doc
c7303e7e66652701891542cceb9f9320  /home/candra/test_1.doc
candra@candra-VirtualBox:~$ sha256sum ~/test_1.doc
96658949c0e94097feac28496c25042d75194ef60772c0bb3a64adb839fd797f  /home/candra/test_1.doc
candra@candra-VirtualBox:~$ sha512sum ~/test_1.doc
bb9511a3308695b8282ce304b0a4959cf2be418e2c6c98a9afb360440fb87944dff6cdb8414f340
f38446ce4fe7e5904150587db87c68683c3036a0b04c5fdda  /home/candra/test_1.doc
candra@candra-VirtualBox:~$ libreoffice ~/test_1.doc
candra@candra-VirtualBox:~$ md5sum ~/test_1.doc
b87539ad8e305047b15c51f9f8fe8d5c  /home/candra/test_1.doc
candra@candra-VirtualBox:~$ sha256sum ~/test_1.doc
67de4ccdec1cc7f9cf3d108f5d27fb882f8b458b316552aa8c720d9b5ecc83f4  /home/candra/test_1.doc
candra@candra-VirtualBox:~$ sha512sum ~/test_1.doc
478d812b110b88f08e18c8f9869ce19508a16e00c5f929917a297022108b6d589c1ae7dd93e6a92
fba437306e779c7f37123e1aed2e1d2e2fb3e47a80c76b3de  /home/candra/test_1.doc
candra@candra-VirtualBox:~$
```

E. Analisis

Hash berubah walaupun isi teks kembali sama karena file DOC menyimpan metadata seperti waktu modifikasi, jumlah penyimpanan, dan informasi revisi yang ikut berubah saat file disimpan.

4. Konfidensialitas dengan EncFS

A. Install EncFS

```
sudo apt update  
sudo apt install encfs
```

B. Buat folder

```
mkdir -p ~/modul15/enc ~/modul15/dec
```

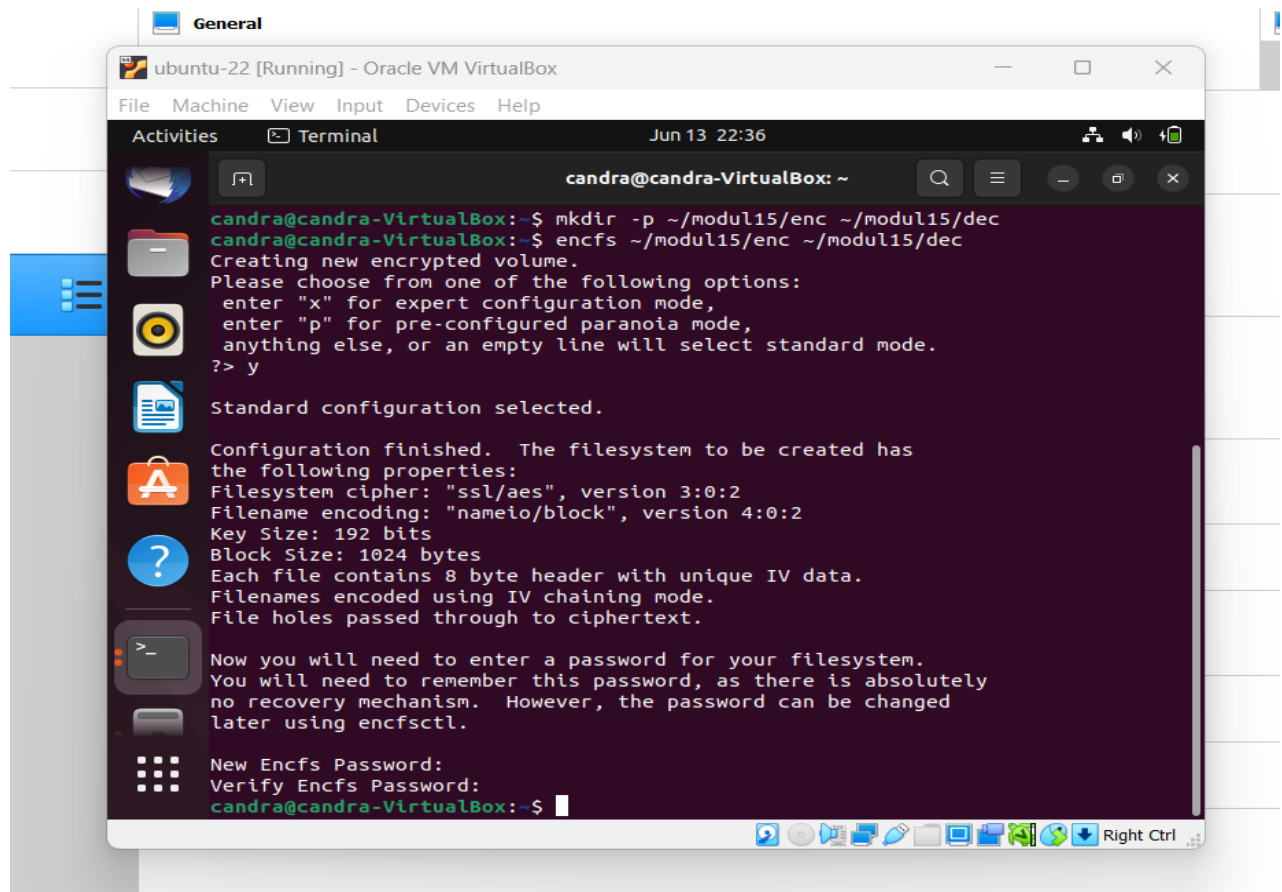
C. Jalankan EncFS

```
encfs ~/modul15/enc ~/modul15/dec
```

Pilih:

y

Kemudian buat password.



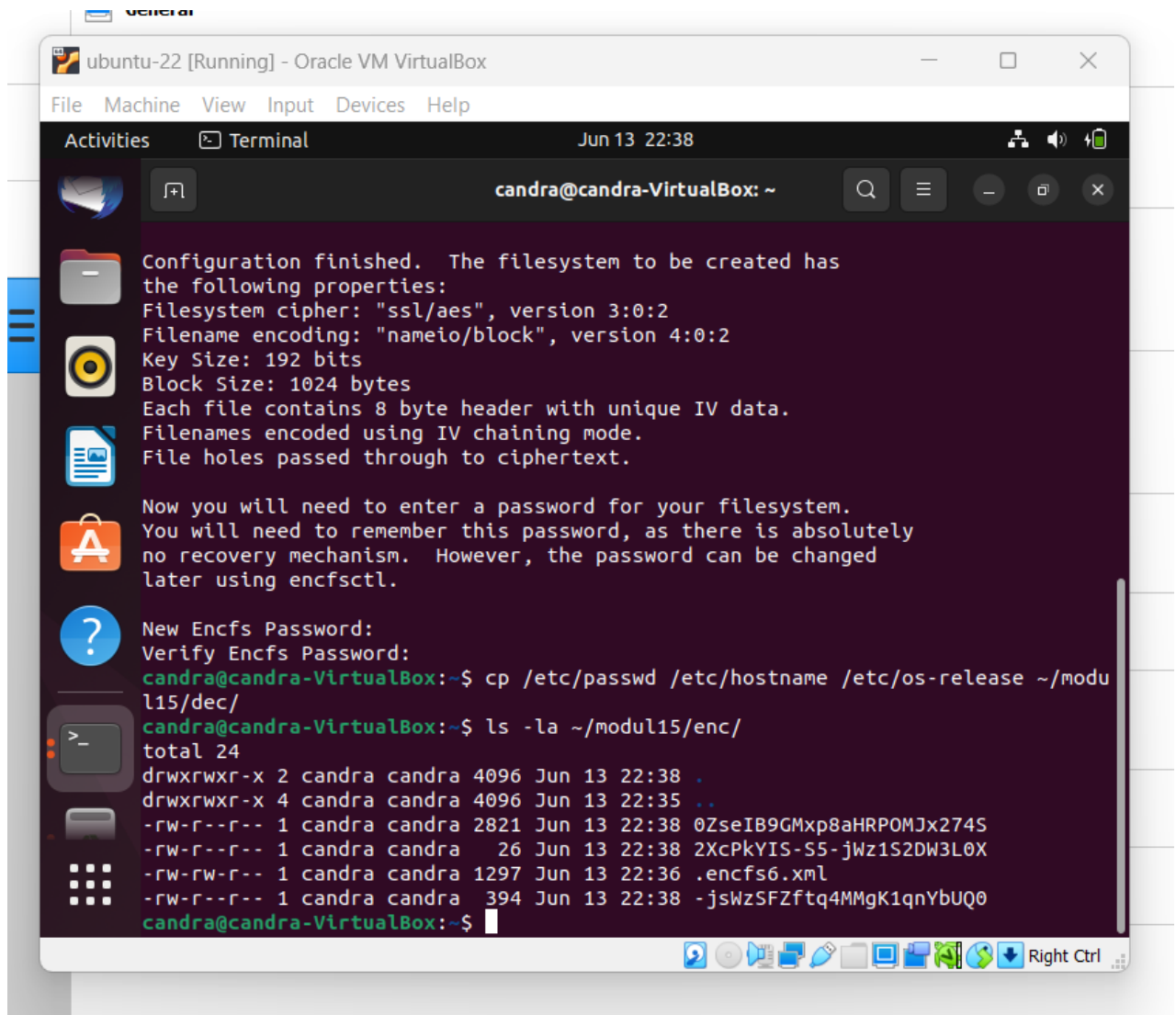
D. Copy 3 file

```
cp /etc/passwd /etc/hostname /etc/os-release ~/modul15/dec/
```

Lihat hasil folder terenkripsi:

```
ls -la ~/modul15/enc/
```

Nama file akan terlihat acak.



E. Hapus file dari enc

Cari salah satu file:

ls ~/modul15/enc

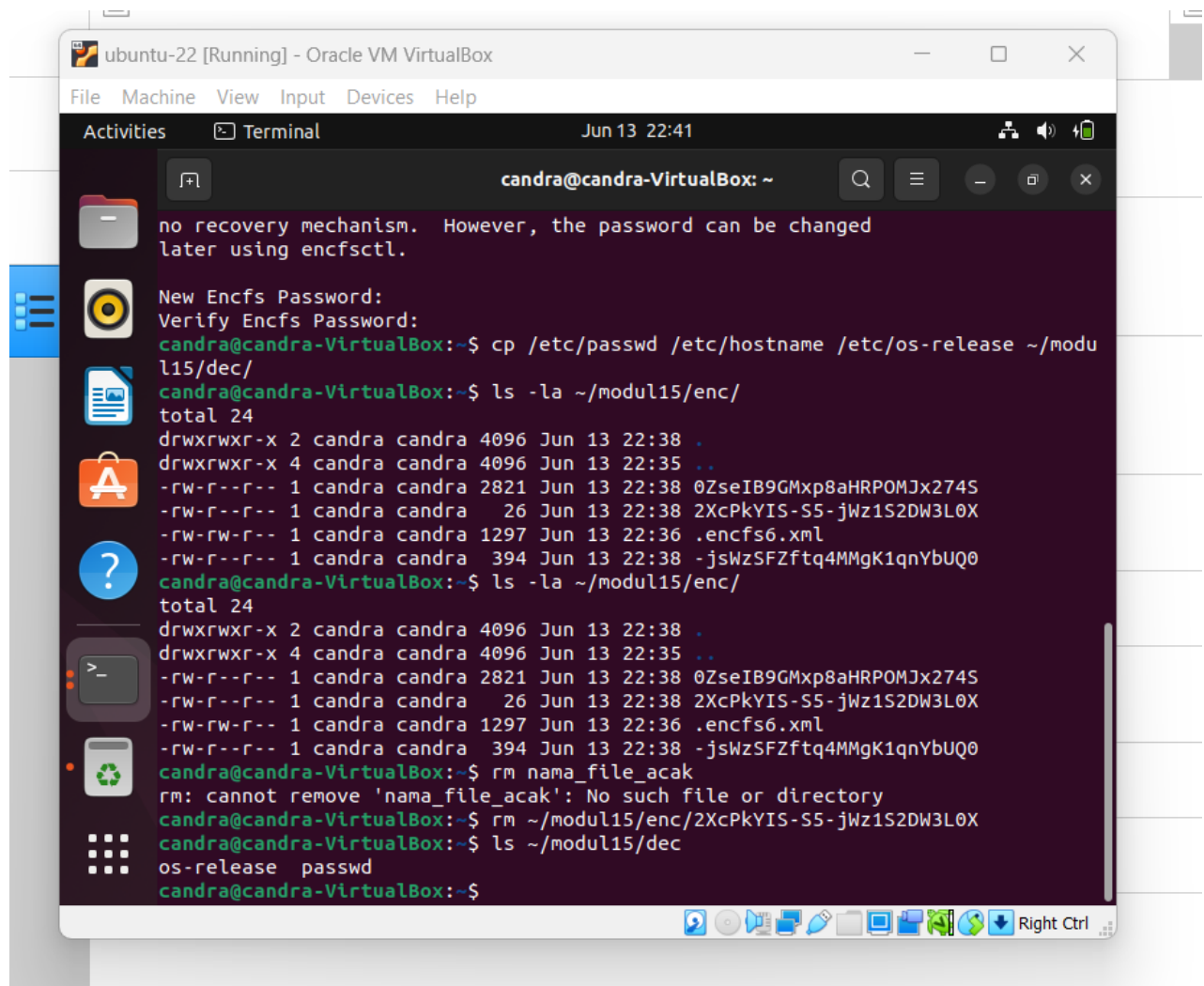
Hapus:

rm ~/modul15/enc/2XcPkYIS-S5-jWz1S2DW3L0X

Cek folder dekripsi:

ls ~/modul15/dec

File yang bersesuaian akan hilang.



The screenshot shows a terminal window titled "ubuntu-22 [Running] - Oracle VM VirtualBox". The terminal output is as follows:

```
no recovery mechanism. However, the password can be changed
later using encfsctl.

New Encfs Password:
Verify Encfs Password:
candra@candra-VirtualBox:~$ cp /etc/passwd /etc/hostname /etc/os-release ~/modu
l15/dec/
candra@candra-VirtualBox:~$ ls -la ~/modul15/enc/
total 24
drwxrwxr-x 2 candra candra 4096 Jun 13 22:38 .
drwxrwxr-x 4 candra candra 4096 Jun 13 22:35 ..
-rw-r--r-- 1 candra candra 2821 Jun 13 22:38 0ZseIB9GMxp8aHRPOMJx274S
-rw-r--r-- 1 candra candra 26 Jun 13 22:38 2XcPkYIS-S5-jWz1S2DW3L0X
-rw-rw-r-- 1 candra candra 1297 Jun 13 22:36 .encfs6.xml
-rw-r--r-- 1 candra candra 394 Jun 13 22:38 -jsWzSFZftq4MMgK1qnYbUQ0
candra@candra-VirtualBox:~$ ls -la ~/modul15/enc/
total 24
drwxrwxr-x 2 candra candra 4096 Jun 13 22:38 .
drwxrwxr-x 4 candra candra 4096 Jun 13 22:35 ..
-rw-r--r-- 1 candra candra 2821 Jun 13 22:38 0ZseIB9GMxp8aHRPOMJx274S
-rw-r--r-- 1 candra candra 26 Jun 13 22:38 2XcPkYIS-S5-jWz1S2DW3L0X
-rw-rw-r-- 1 candra candra 1297 Jun 13 22:36 .encfs6.xml
-rw-r--r-- 1 candra candra 394 Jun 13 22:38 -jsWzSFZftq4MMgK1qnYbUQ0
candra@candra-VirtualBox:~$ rm nama_file_acak
rm: cannot remove 'nama_file_acak': No such file or directory
candra@candra-VirtualBox:~$ rm ~/modul15/enc/2XcPkYIS-S5-jWz1S2DW3L0X
candra@candra-VirtualBox:~$ ls ~/modul15/dec
os-release passwd
candra@candra-VirtualBox:~$
```

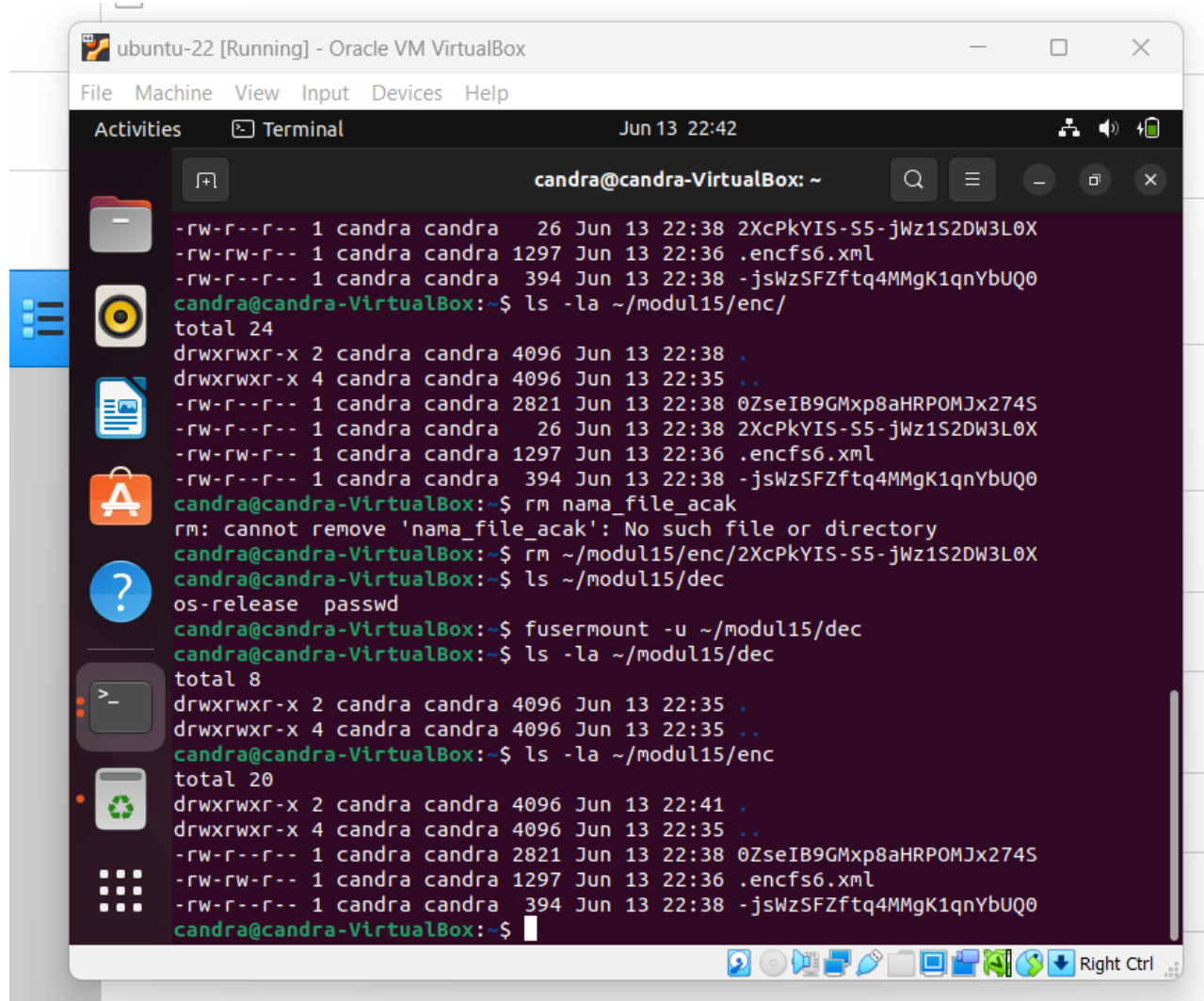
F. Unmount

`fusermount -u ~/modul15/dec`

Cek:

`ls -la ~/modul15/dec`

`ls -la ~/modul15/enc`



The screenshot shows a terminal window titled "ubuntu-22 [Running] - Oracle VM VirtualBox". The terminal output is as follows:

```
candra@candra-VirtualBox: ~  
-rw-r--r-- 1 candra candra 26 Jun 13 22:38 2XcPkYIS-S5-jWz1S2DW3L0X  
-rw-rw-r-- 1 candra candra 1297 Jun 13 22:36 .encfs6.xml  
-rw-r--r-- 1 candra candra 394 Jun 13 22:38 -jsWzSFZftq4MMgK1qnYbUQ0  
candra@candra-VirtualBox:~$ ls -la ~/modul15/enc/  
total 24  
drwxrwxr-x 2 candra candra 4096 Jun 13 22:38 .  
drwxrwxr-x 4 candra candra 4096 Jun 13 22:35 ..  
-rw-r--r-- 1 candra candra 2821 Jun 13 22:38 0ZseIB9GMxp8aHRPOMJx274S  
-rw-r--r-- 1 candra candra 26 Jun 13 22:38 2XcPkYIS-S5-jWz1S2DW3L0X  
-rw-rw-r-- 1 candra candra 1297 Jun 13 22:36 .encfs6.xml  
-rw-r--r-- 1 candra candra 394 Jun 13 22:38 -jsWzSFZftq4MMgK1qnYbUQ0  
candra@candra-VirtualBox:~$ rm nama_file_acak  
rm: cannot remove 'nama_file_acak': No such file or directory  
candra@candra-VirtualBox:~$ rm ~/modul15/enc/2XcPkYIS-S5-jWz1S2DW3L0X  
candra@candra-VirtualBox:~$ ls ~/modul15/dec  
os-release passwd  
candra@candra-VirtualBox:~$ fusermount -u ~/modul15/dec  
candra@candra-VirtualBox:~$ ls -la ~/modul15/dec  
total 8  
drwxrwxr-x 2 candra candra 4096 Jun 13 22:35 .  
drwxrwxr-x 4 candra candra 4096 Jun 13 22:35 ..  
candra@candra-VirtualBox:~$ ls -la ~/modul15/enc  
total 20  
drwxrwxr-x 2 candra candra 4096 Jun 13 22:41 .  
drwxrwxr-x 4 candra candra 4096 Jun 13 22:35 ..  
-rw-r--r-- 1 candra candra 2821 Jun 13 22:38 0ZseIB9GMxp8aHRPOMJx274S  
-rw-rw-r-- 1 candra candra 1297 Jun 13 22:36 .encfs6.xml  
-rw-r--r-- 1 candra candra 394 Jun 13 22:38 -jsWzSFZftq4MMgK1qnYbUQ0  
candra@candra-VirtualBox:~$
```

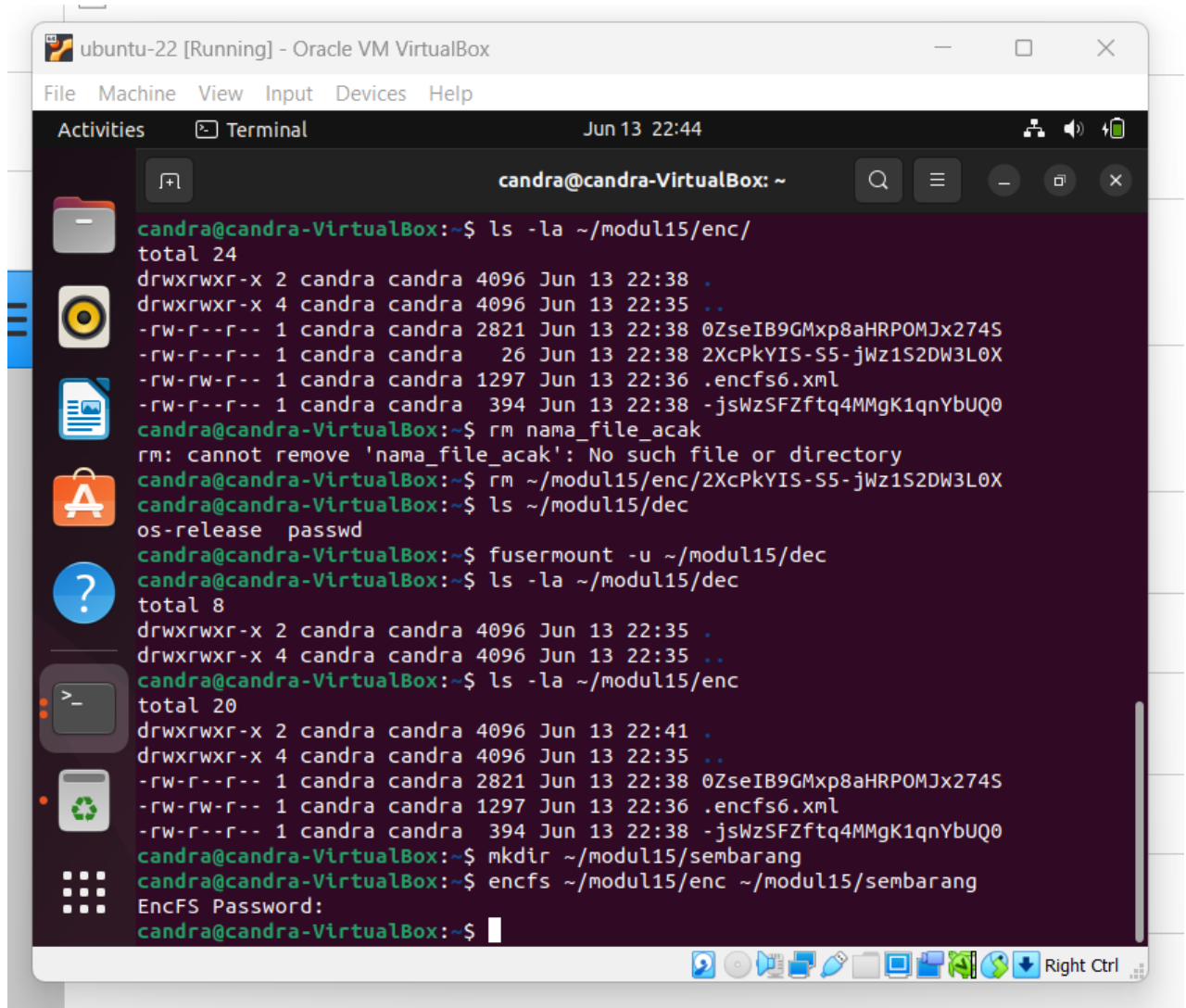
G. Mount ke folder lain

`mkdir ~/modul15/sembarang`

`encfs ~/modul15/enc ~/modul15/sembarang`

Masukkan password yang sama.

File akan terbaca normal kembali.



```
ubuntu-22 [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help
Activities Terminal Jun 13 22:44
candra@candra-VirtualBox: ~

candra@candra-VirtualBox:~$ ls -la ~/modul15/enc/
total 24
drwxrwxr-x 2 candra candra 4096 Jun 13 22:38 .
drwxrwxr-x 4 candra candra 4096 Jun 13 22:35 ..
-rw-r--r-- 1 candra candra 2821 Jun 13 22:38 0ZseIB9GMxp8aHRPOMJx274S
-rw-r--r-- 1 candra candra 26 Jun 13 22:38 2XcPkYIS-S5-jWz1S2DW3L0X
-rw-rw-r-- 1 candra candra 1297 Jun 13 22:36 .encfs6.xml
-rw-r--r-- 1 candra candra 394 Jun 13 22:38 -jsWzSFZftq4MMgK1qnYbUQ0
candra@candra-VirtualBox:~$ rm nama_file_acak
rm: cannot remove 'nama_file_acak': No such file or directory
candra@candra-VirtualBox:~$ rm ~/modul15/enc/2XcPkYIS-S5-jWz1S2DW3L0X
candra@candra-VirtualBox:~$ ls ~/modul15/dec
os-release passwd
candra@candra-VirtualBox:~$ fusermount -u ~/modul15/dec
candra@candra-VirtualBox:~$ ls -la ~/modul15/dec
total 8
drwxrwxr-x 2 candra candra 4096 Jun 13 22:35 .
drwxrwxr-x 4 candra candra 4096 Jun 13 22:35 ..
candra@candra-VirtualBox:~$ ls -la ~/modul15/enc
total 20
drwxrwxr-x 2 candra candra 4096 Jun 13 22:41 .
drwxrwxr-x 4 candra candra 4096 Jun 13 22:35 ..
-rw-r--r-- 1 candra candra 2821 Jun 13 22:38 0ZseIB9GMxp8aHRPOMJx274S
-rw-rw-r-- 1 candra candra 1297 Jun 13 22:36 .encfs6.xml
-rw-r--r-- 1 candra candra 394 Jun 13 22:38 -jsWzSFZftq4MMgK1qnYbUQ0
candra@candra-VirtualBox:~$ mkdir ~/modul15/sembarang
candra@candra-VirtualBox:~$ encfs ~/modul15/enc ~/modul15/sembarang
EncFS Password:
candra@candra-VirtualBox:~$
```

5. GPG (GNU Privacy Guard)

A. Install GPG

sudo apt update

sudo apt install gnupg

B. Membuat Key Pair

gpg --gen-key

Isi:

Nama

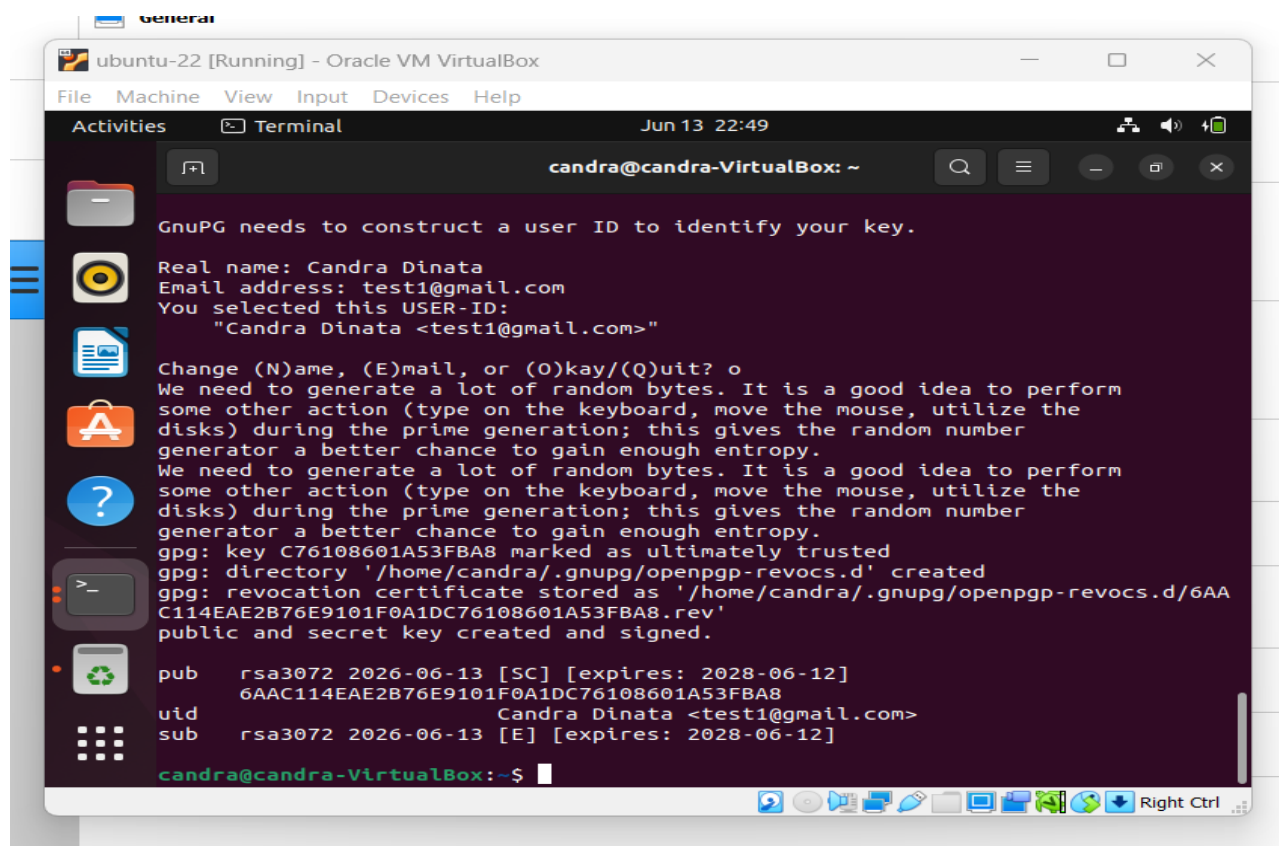
Email

Password

Contoh:

Nama : Candra Dinata

Email : test1@gmail.com



The screenshot shows a terminal window titled "ubuntu-22 [Running] - Oracle VM VirtualBox". The terminal output is as follows:

```
gpg --gen-key

GnuPG needs to construct a user ID to identify your key.

Real name: Candra Dinata
Email address: test1@gmail.com
You selected this USER-ID:
    "Candra Dinata <test1@gmail.com>"

Change (N)ame, (E)mail, or (O)kay/(Q)uit? o
We need to generate a lot of random bytes. It is a good idea to perform
some other action (type on the keyboard, move the mouse, utilize the
disks) during the prime generation; this gives the random number
generator a better chance to gain enough entropy.
We need to generate a lot of random bytes. It is a good idea to perform
some other action (type on the keyboard, move the mouse, utilize the
disks) during the prime generation; this gives the random number
generator a better chance to gain enough entropy.
gpg: key C76108601A53FBA8 marked as ultimately trusted
gpg: directory '/home/candra/.gnupg/openpgp-revocs.d' created
gpg: revocation certificate stored as '/home/candra/.gnupg/openpgp-revocs.d/6AA
C114EAE2B76E9101F0A1DC76108601A53FBA8.rev'
public and secret key created and signed.

pub   rsa3072 2026-06-13 [SC] [expires: 2028-06-12]
      6AAC114EAE2B76E9101F0A1DC76108601A53FBA8
uid           Candra Dinata <test1@gmail.com>
sub   rsa3072 2026-06-13 [E] [expires: 2028-06-12]

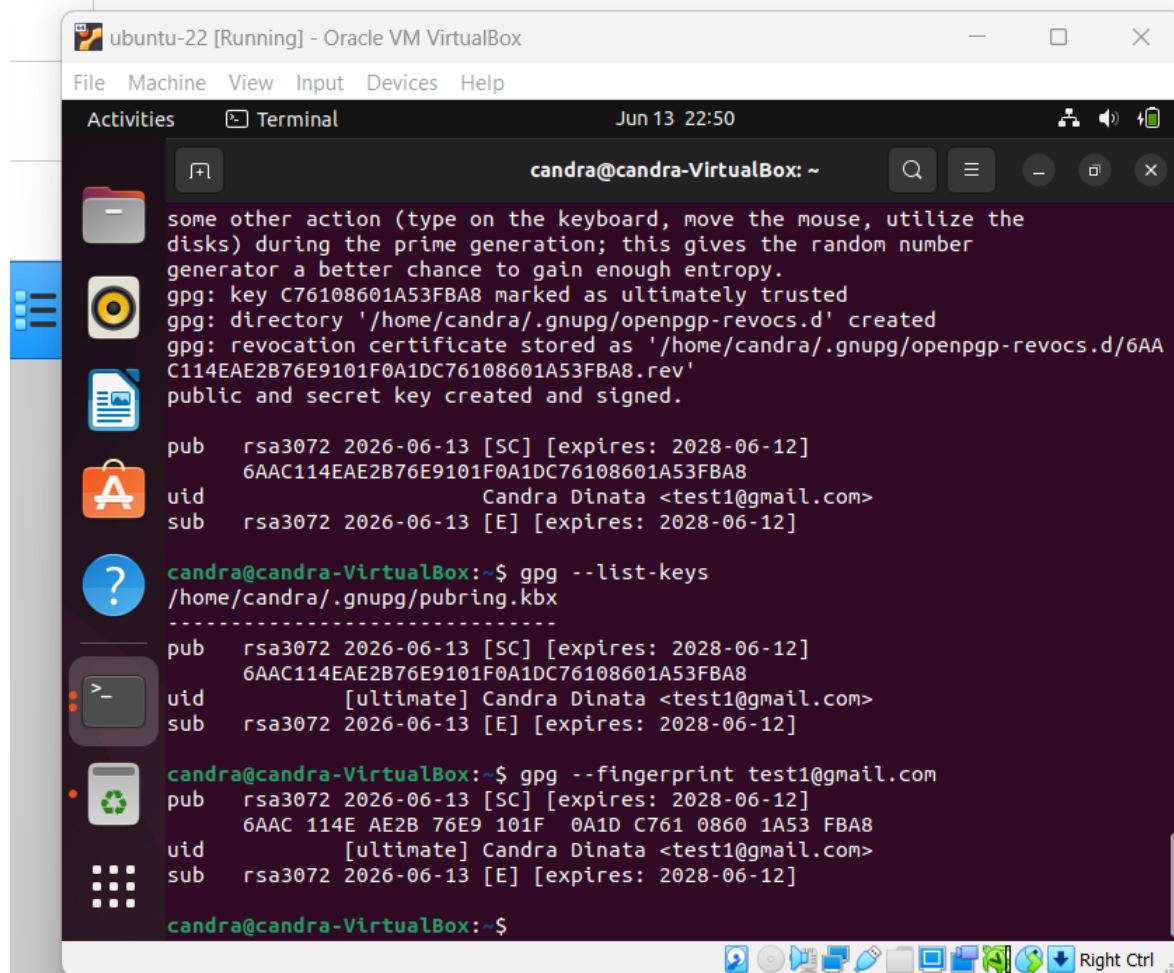
candra@candra-VirtualBox: ~$
```

C. Melihat Key

gpg --list-keys

Melihat fingerprint:

gpg --fingerprint test1@gmail.com



The screenshot shows a terminal window titled 'candra@candra-VirtualBox: ~' with a dark background. The window displays the output of several GPG commands. It starts with a message about entropy generation, followed by the creation of a key for 'test1@gmail.com'. The key is listed as 'rsa3072 2026-06-13 [SC] [expires: 2028-06-12]' with a fingerprint of '6AAC114EAE2B76E9101F0A1DC76108601A53FBA8'. The user 'Candra Dinata' is associated with the key. The terminal also shows the command 'gpg --list-keys' and 'gpg --fingerprint test1@gmail.com', both of which output the same key information.

```
candra@candra-VirtualBox: ~  
some other action (type on the keyboard, move the mouse, utilize the  
disks) during the prime generation; this gives the random number  
generator a better chance to gain enough entropy.  
gpg: key C76108601A53FBA8 marked as ultimately trusted  
gpg: directory '/home/candra/.gnupg/openpgp-revocs.d' created  
gpg: revocation certificate stored as '/home/candra/.gnupg/openpgp-revocs.d/6AA  
C114EAE2B76E9101F0A1DC76108601A53FBA8.rev'  
public and secret key created and signed.  
  
pub   rsa3072 2026-06-13 [SC] [expires: 2028-06-12]  
      6AAC114EAE2B76E9101F0A1DC76108601A53FBA8  
uid           Candra Dinata <test1@gmail.com>  
sub   rsa3072 2026-06-13 [E] [expires: 2028-06-12]  
  
candra@candra-VirtualBox:~$ gpg --list-keys  
/home/candra/.gnupg/pubring.kbx  
-----  
pub   rsa3072 2026-06-13 [SC] [expires: 2028-06-12]  
      6AAC114EAE2B76E9101F0A1DC76108601A53FBA8  
uid           [ultimate] Candra Dinata <test1@gmail.com>  
sub   rsa3072 2026-06-13 [E] [expires: 2028-06-12]  
  
candra@candra-VirtualBox:~$ gpg --fingerprint test1@gmail.com  
pub   rsa3072 2026-06-13 [SC] [expires: 2028-06-12]  
      6AAC 114E AE2B 76E9 101F 0A1D C761 0860 1A53 FBA8  
uid           [ultimate] Candra Dinata <test1@gmail.com>  
sub   rsa3072 2026-06-13 [E] [expires: 2028-06-12]  
  
candra@candra-VirtualBox:~$
```

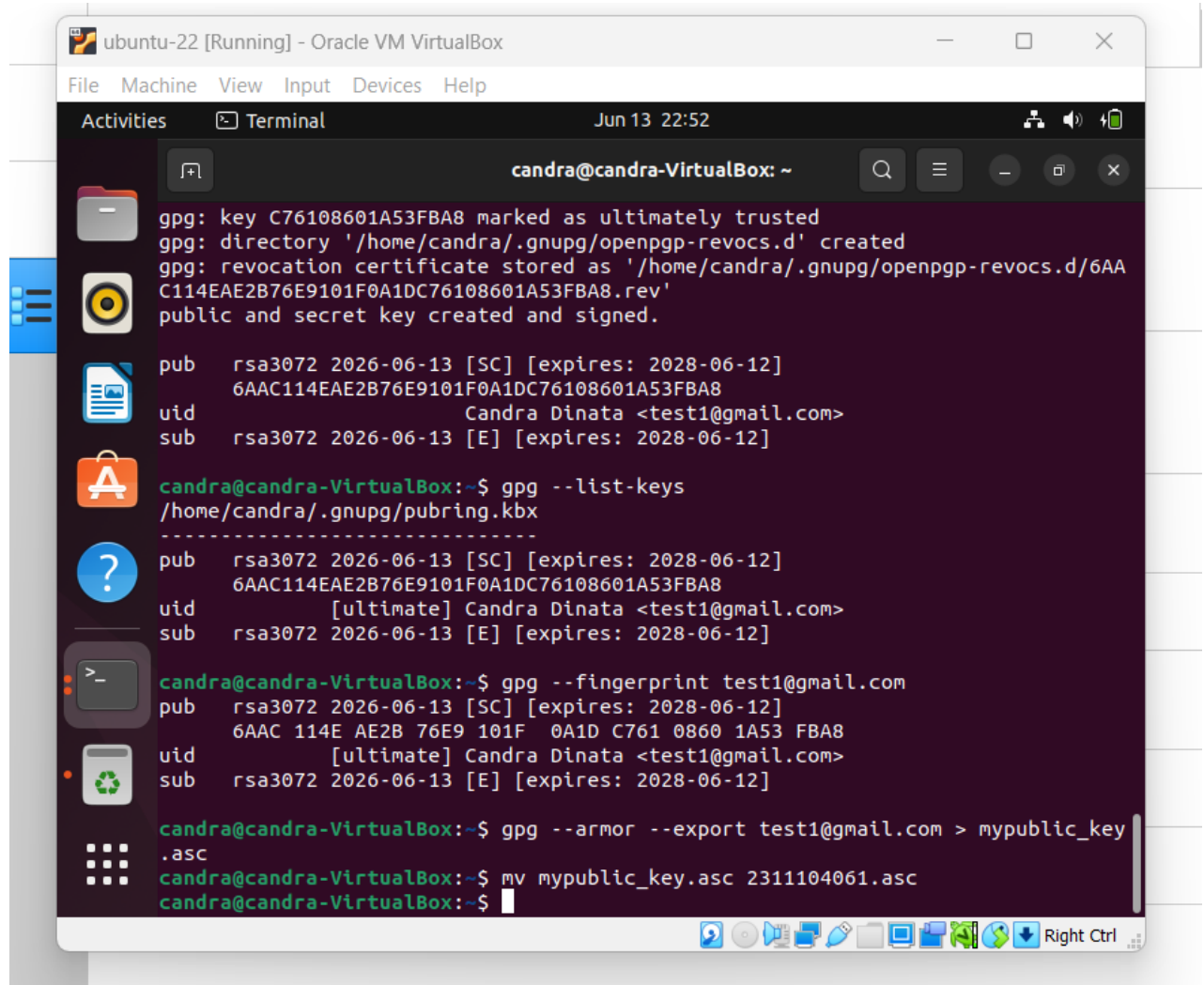
D. Export Public Key

Ganti dengan email milikmu:

```
gpg --armor --export test1@gmail.com > mypublic_key.asc
```

Rename sesuai NIM:

```
mv mypublic_key.asc 2311104061.asc
```



The screenshot shows a terminal window titled 'candra@candra-VirtualBox: ~' with the following commands and output:

```
gpg: key C76108601A53FBA8 marked as ultimately trusted
gpg: directory '/home/candra/.gnupg/openpgp-revocs.d' created
gpg: revocation certificate stored as '/home/candra/.gnupg/openpgp-revocs.d/6AAC114EAE2B76E9101F0A1DC76108601A53FBA8.rev'
public and secret key created and signed.

pub  rsa3072 2026-06-13 [SC] [expires: 2028-06-12]
     6AAC114EAE2B76E9101F0A1DC76108601A53FBA8
uid  Candra Dinata <test1@gmail.com>
sub  rsa3072 2026-06-13 [E] [expires: 2028-06-12]

candra@candra-VirtualBox:~$ gpg --list-keys /home/candra/.gnupg/pubring.kbx
-----
pub  rsa3072 2026-06-13 [SC] [expires: 2028-06-12]
     6AAC114EAE2B76E9101F0A1DC76108601A53FBA8
uid  [ultimate] Candra Dinata <test1@gmail.com>
sub  rsa3072 2026-06-13 [E] [expires: 2028-06-12]

candra@candra-VirtualBox:~$ gpg --fingerprint test1@gmail.com
pub  rsa3072 2026-06-13 [SC] [expires: 2028-06-12]
     6AAC 114E AE2B 76E9 101F 0A1D C761 0860 1A53 FBA8
uid  [ultimate] Candra Dinata <test1@gmail.com>
sub  rsa3072 2026-06-13 [E] [expires: 2028-06-12]

candra@candra-VirtualBox:~$ gpg --armor --export test1@gmail.com > mypublic_key.asc
candra@candra-VirtualBox:~$ mv mypublic_key.asc 2311104061.asc
candra@candra-VirtualBox:~$
```

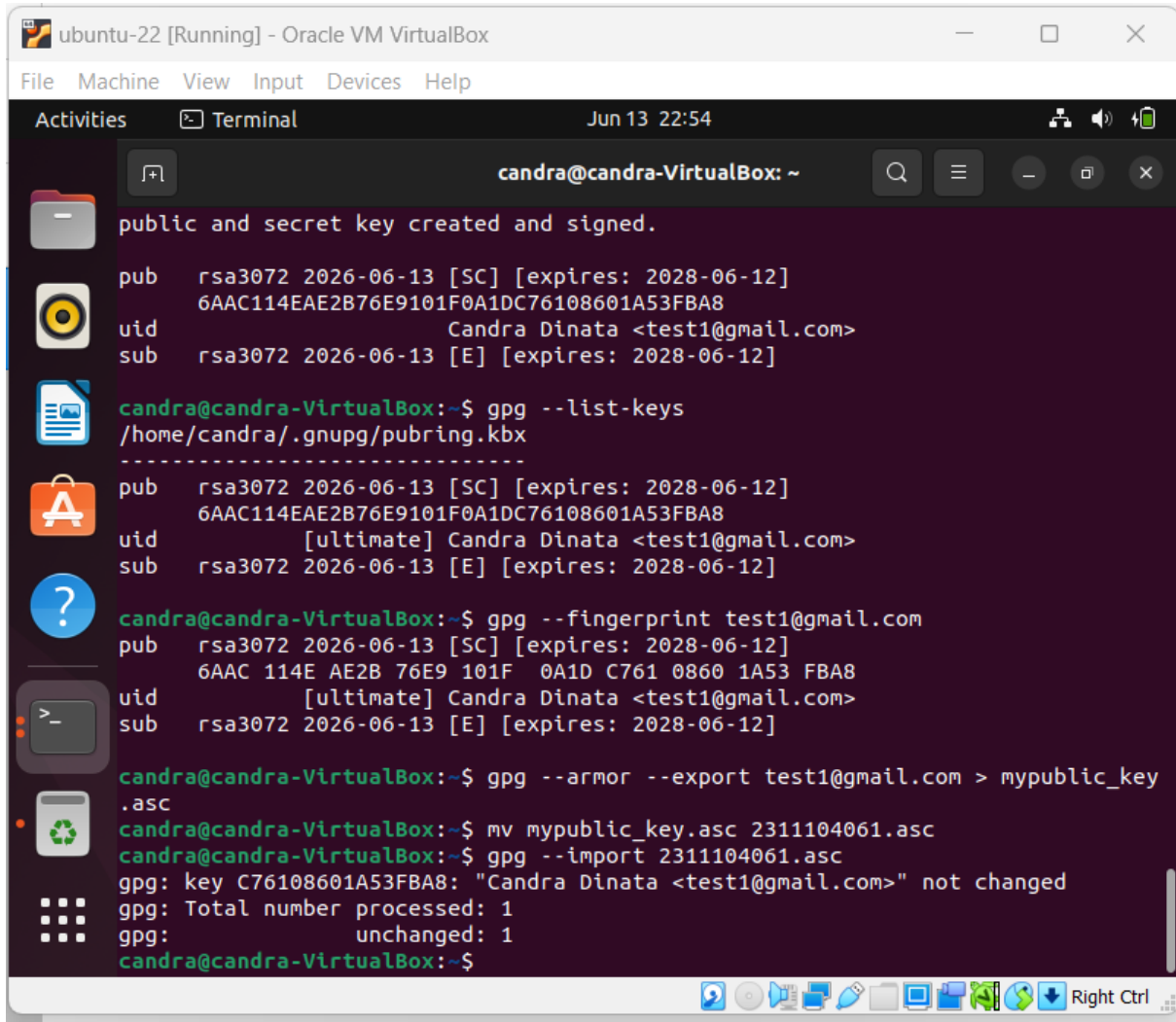
E. Import Public Key Teman

Misal teman memberi file:

2311104061.asc

Import:

`gpg --import 2311104061.asc`



```
ubuntu-22 [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help
Activities Terminal Jun 13 22:54
candra@candra-VirtualBox: ~
public and secret key created and signed.
pub  rsa3072 2026-06-13 [SC] [expires: 2028-06-12]
    6AAC114EAE2B76E9101F0A1DC76108601A53FBA8
uid                               Candra Dinata <test1@gmail.com>
sub  rsa3072 2026-06-13 [E] [expires: 2028-06-12]

candra@candra-VirtualBox:~$ gpg --list-keys
/home/candra/.gnupg/pubring.kbx
-----
pub  rsa3072 2026-06-13 [SC] [expires: 2028-06-12]
    6AAC114EAE2B76E9101F0A1DC76108601A53FBA8
uid                               [ultimate] Candra Dinata <test1@gmail.com>
sub  rsa3072 2026-06-13 [E] [expires: 2028-06-12]

candra@candra-VirtualBox:~$ gpg --fingerprint test1@gmail.com
pub  rsa3072 2026-06-13 [SC] [expires: 2028-06-12]
    6AAC 114E AE2B 76E9 101F 0A1D C761 0860 1A53 FBA8
uid                               [ultimate] Candra Dinata <test1@gmail.com>
sub  rsa3072 2026-06-13 [E] [expires: 2028-06-12]

candra@candra-VirtualBox:~$ gpg --armor --export test1@gmail.com > mypublic_key
.asc
candra@candra-VirtualBox:~$ mv mypublic_key.asc 2311104061.asc
candra@candra-VirtualBox:~$ gpg --import 2311104061.asc
gpg: key C76108601A53FBA8: "Candra Dinata <test1@gmail.com>" not changed
gpg: Total number processed: 1
gpg:      unchanged: 1
candra@candra-VirtualBox:~$
```

F. Enkripsi Pesan

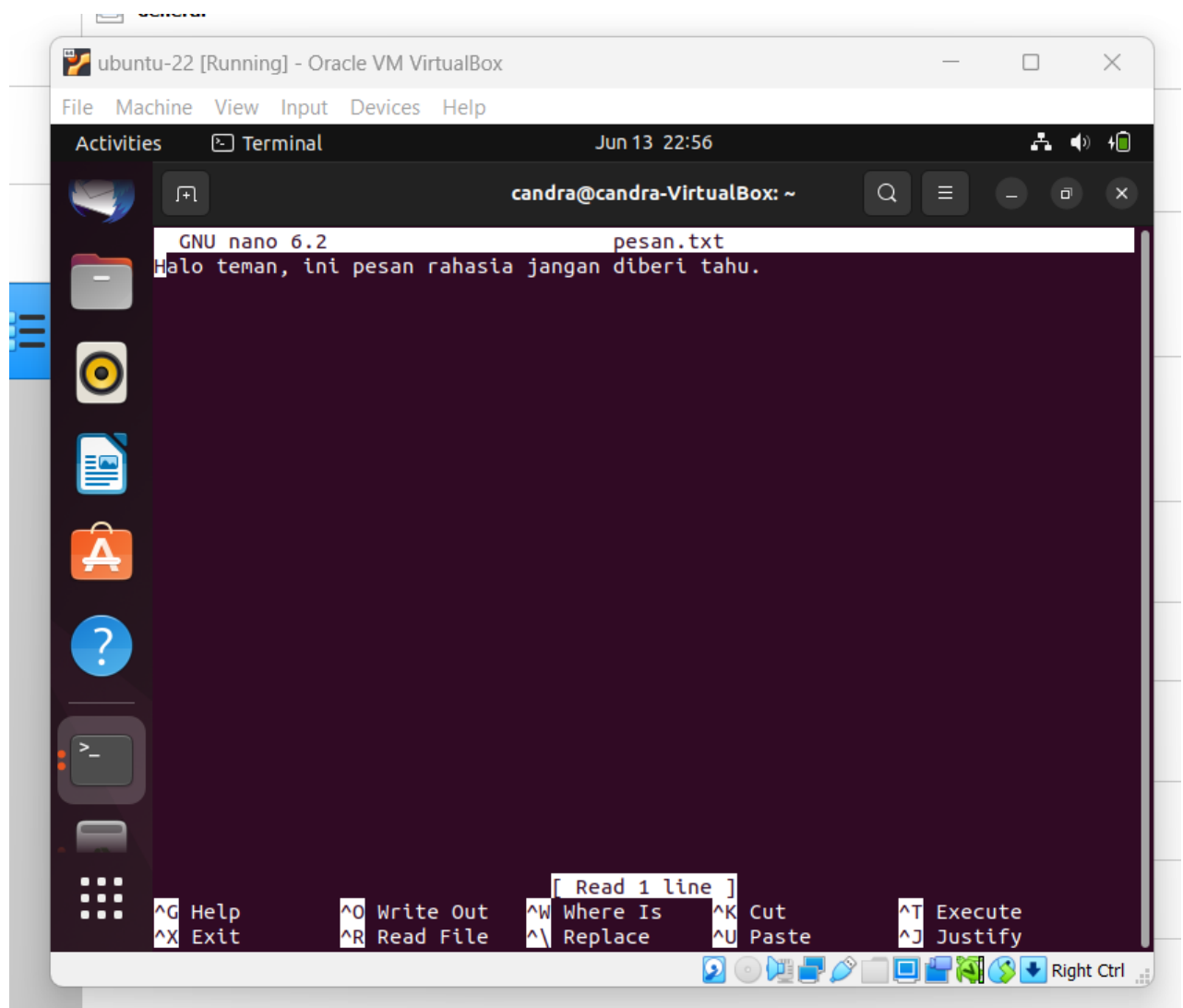
Buat file:

nano pesan.txt

Isi:

Halo teman, ini pesan rahasia jangan diberi tahu.

Simpan.

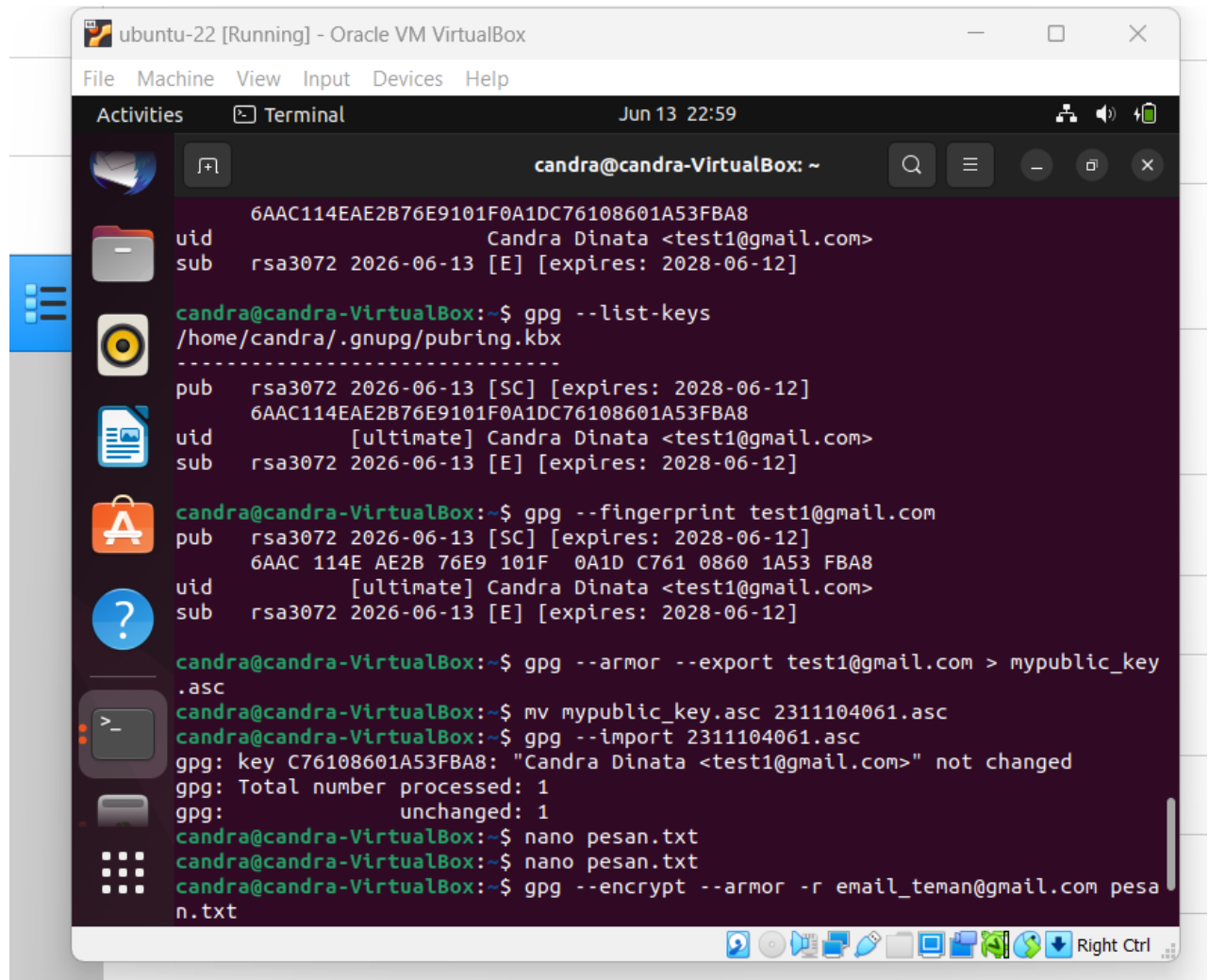


Enkripsi:

gpg --encrypt --armor -r email_teman@gmail.com pesan.txt

Hasil:

pesan.txt.asc



```
ubuntu-22 [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help
Activities Terminal Jun 13 22:59
candra@candra-VirtualBox: ~

6AAC114EAE2B76E9101F0A1DC76108601A53FBA8
uid          Candra Dinata <test1@gmail.com>
sub rsa3072 2026-06-13 [E] [expires: 2028-06-12]

candra@candra-VirtualBox:~$ gpg --list-keys
/home/candra/.gnupg/pubring.kbx
-----
pub rsa3072 2026-06-13 [SC] [expires: 2028-06-12]
6AAC114EAE2B76E9101F0A1DC76108601A53FBA8
uid          [ultimate] Candra Dinata <test1@gmail.com>
sub rsa3072 2026-06-13 [E] [expires: 2028-06-12]

candra@candra-VirtualBox:~$ gpg --fingerprint test1@gmail.com
pub rsa3072 2026-06-13 [SC] [expires: 2028-06-12]
6AAC 114E AE2B 76E9 101F 0A1D C761 0860 1A53 FBA8
uid          [ultimate] Candra Dinata <test1@gmail.com>
sub rsa3072 2026-06-13 [E] [expires: 2028-06-12]

candra@candra-VirtualBox:~$ gpg --armor --export test1@gmail.com > mypublic_key
.asc
candra@candra-VirtualBox:~$ mv mypublic_key.asc 2311104061.asc
candra@candra-VirtualBox:~$ gpg --import 2311104061.asc
gpg: key C76108601A53FBA8: "Candra Dinata <test1@gmail.com>" not changed
gpg: Total number processed: 1
gpg:      unchanged: 1

candra@candra-VirtualBox:~$ nano pesan.txt
candra@candra-VirtualBox:~$ nano pesan.txt
candra@candra-VirtualBox:~$ gpg --encrypt --armor -r email_teman@gmail.com pesa
n.txt
```

G. Dekripsi Pesan

Jika menerima file:

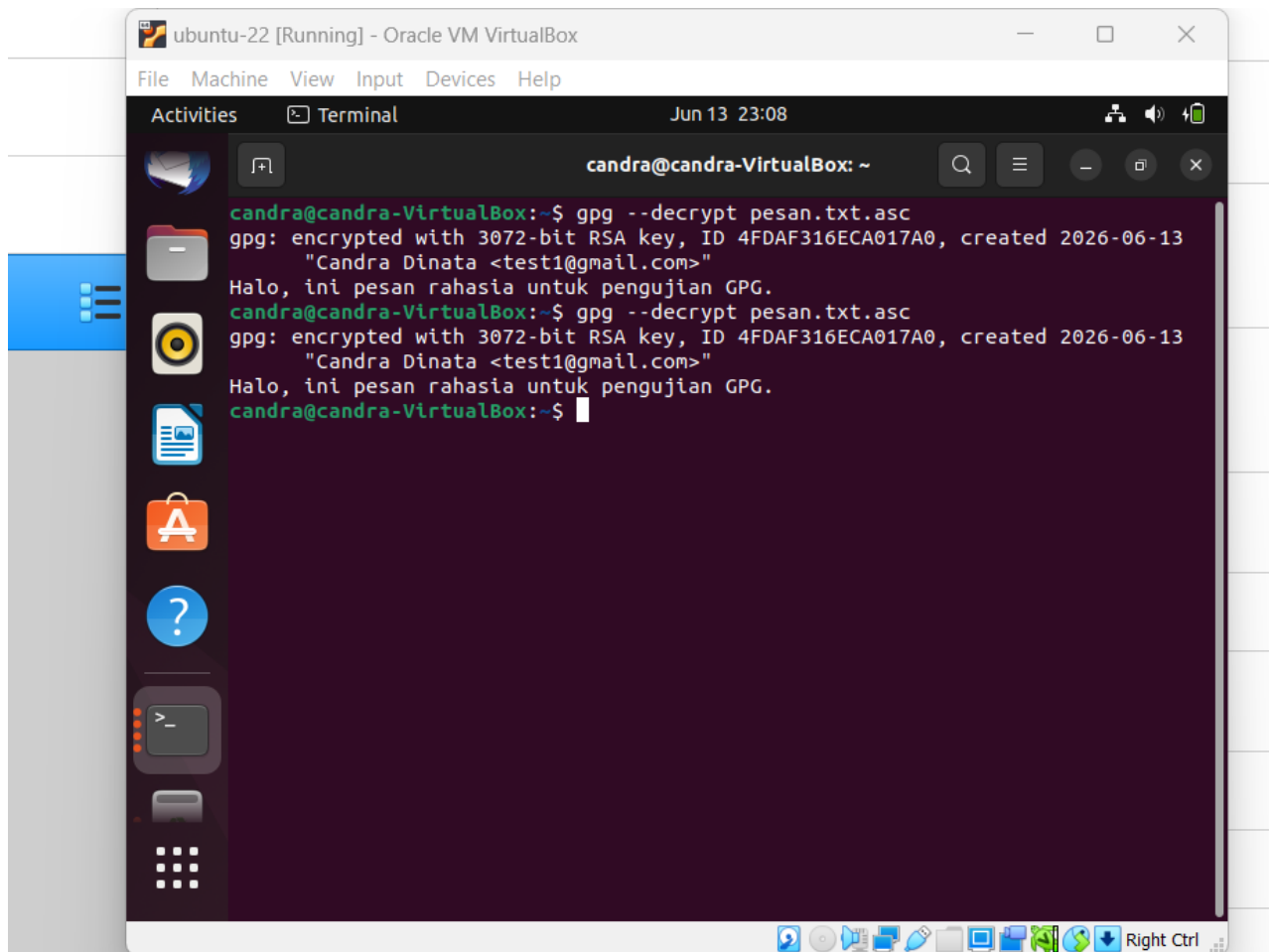
pesan.txt.asc

Jalankan:

`gpg --decrypt pesan.txt.asc`

atau

gpg --output hasil.txt --decrypt pesan.txt.asc



The screenshot shows a terminal window titled "ubuntu-22 [Running] - Oracle VM VirtualBox". The window has a menu bar with "File", "Machine", "View", "Input", "Devices", and "Help". Below the menu bar is a toolbar with "Activities" and "Terminal" buttons, and a clock showing "Jun 13 23:08". The terminal itself has a title bar "candra@candra-VirtualBox: ~" and a search bar. The terminal output shows the command `gpg --decrypt pesan.txt.asc` being executed twice. The output for each execution is: `gpg: encrypted with 3072-bit RSA key, ID 4FDAF316ECA017A0, created 2026-06-13`, `"Candra Dinata <test1@gmail.com>"`, and `Halo, ini pesan rahasia untuk pengujian GPG.`. The terminal has a dark purple background and a light blue cursor. On the left side of the terminal window, there is a vertical sidebar with icons for a file manager, a terminal, a music player, a document, an application store, a help icon, and a grid of application icons. At the bottom of the terminal window, there is a taskbar with various system icons and a "Right Ctrl" button.

```
candra@candra-VirtualBox:~$ gpg --decrypt pesan.txt.asc
gpg: encrypted with 3072-bit RSA key, ID 4FDAF316ECA017A0, created 2026-06-13
"Candra Dinata <test1@gmail.com>"
Halo, ini pesan rahasia untuk pengujian GPG.
candra@candra-VirtualBox:~$ gpg --decrypt pesan.txt.asc
gpg: encrypted with 3072-bit RSA key, ID 4FDAF316ECA017A0, created 2026-06-13
"Candra Dinata <test1@gmail.com>"
Halo, ini pesan rahasia untuk pengujian GPG.
candra@candra-VirtualBox:~$
```